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MASTER THESIS

Exploring the Advantages of Multi-View Satellite Image Data for Robust Cloud Segmentation

submitted by
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Abstract

Cloud and cloud shadow detection is a fundamental pre-processing step in remote sensing, yet it remains a persistent challenge due to the inherent limitations of optical satellite imagery. With approximately 67% of the Earth's surface covered by clouds, accurate surface observation is frequently obstructed, hindering applications ranging from crop monitoring to disaster response. Traditional segmentation algorithms, which rely primarily on optical spectral features, often fail in spectrally complex environments, such as distinguishing bright clouds from snow in polar regions and lack resilience against sensor data loss.

This thesis explores the advantages of multi-view deep learning to address these robustness challenges. We propose and evaluate a multi-modal U-Net framework that fuses Sentinel-2 multispectral optical data with Sentinel-1 Synthetic Aperture Radar (SAR), Digital Elevation Models (DEM), Water Occurrence [44], Azimuth and Land Cover data [57]. To investigate optimal fusion strategies, we implement and compare a baseline Concatenation Fusion with an adaptive Gated Fusion [4], which dynamically weights feature importance.

The proposed models are validated on the global CloudSEN12 [5] benchmark dataset, assessing performance across diverse climatic conditions, cloud cover percentages, land cover types and difficulty levels. Experimental results demonstrate that multi-view integration significantly outperforms single-view optical baselines, particularly in "hard-to-classify" scenarios. Specifically, the inclusion of SAR and DEM successfully disentangles the spectral confusion between clouds and ice in polar and alpine regions. While the Concatenation model achieved the highest overall quantitative metrics, the Gated Fusion model demonstrated superior consistency and stability across varying environmental conditions. Furthermore, robustness experiments simulating band dropout reveal that while multi-view fusion provides some resilience, specific training interventions are required to fully mitigate performance degradation during severe sensor failures. Ultimately, this work establishes that leveraging heterogeneous satellite data is a critical pathway toward achieving truly robust, global cloud and cloud shadow segmentation.

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Chapter 1
Introduction

This master's thesis investigates the efficacy of multi-view learning for segmenting clouds and cloud shadows in satellite images. It further investigates whether gated fusion, as an adaptive learning strategy, is actually resilient to information-loss scenarios, such as band dropout.

1.1 Motivation

Remote sensing is the acquisition of information about the Earth's surface and atmosphere by analyzing data collected by sensors that are not in physical contact with the object or area being observed. These sensors can be mounted on various platforms, including satellites, aircraft, and drones. Remote sensing is used in a wide range of applications, including land-use and land-cover mapping, natural resource management, disaster response, and environmental monitoring. The European Space Agency's Copernicus program, particularly the Sentinel-2 mission, provides high-resolution optical satellite imagery almost daily, which forms the backbone of these applications. However, the amount of usable data that is generated is fundamentally constrained by atmospheric conditions. Global estimates suggest that approximately 67% of the Earth's surface is covered by clouds at any given time [23].

For passive optical sensors that rely solely on solar reflectance for data, cloud cover poses a significant threat, leading to blind pixels where terrestrial information is partially or completely occluded. Consequently, cloud and cloud shadow segmentation is not merely a technical computer vision task; it is the critical pre-processing step required to validate data before any downstream analysis can occur. Failure to accurately segment clouds results in significant data loss or the propagation of noise into derived products, e.g., classifying a cloud as snow or vice versa. By analysing the reflected sunlight from the Earth's surface, remote sensing can provide valuable insights into the physical, biological, and chemical properties of the target area, making it a powerful tool for Earth observation and analysis. Clouds are an integral part of Earth's atmosphere and the Earth's water cycle, which maintains life as we know it. Clouds come in various shapes and sizes, posing a significant hindrance to scientists observing the Earth's surface from satellites.

This is because most satellites have optical sensors onboard that rely on light reflected from Earth's surface. Clouds and cloud shadows obscure the optical information one might need by reflecting sunlight before it reaches the Earth's surface.

1.2 Problem Statement and Research Goals

1.3 Research Objectives

The main objective of this thesis is to develop and evaluate a robust, multi-view deep learning system that effectively fuses heterogeneous satellite data (Optical, SAR, DEM, and other features) to improve cloud and cloud shadow segmentation in spectrally complex settings and to be resilient against information loss.

To achieve this, the following sub-objectives are defined:

1. **Sub-Objective 1 (Fusion Strategy)**: To design and validate an adaptive Gated Fusion approach that can dynamically weigh the importance of Sentinel-1, Sentinel-2 and other auxiliary features, lessening the "weakest link" overfitting problem identified in multi-modal learning.

2. **Sub-Objective 2 (Representation Learning)**: To examine the efficacy of Pretext Task Learning by developing an "Inverted U-Net" architecture. This objective aims to determine if an optical-only model can learn to reconstruct geometric modalities (SAR, DEM and Landcover), thereby enforcing a physics-aware understanding of the scene without requiring multi-modal inputs at inference time.

3. **Sub-Objective 3 (Global Validation)**: To thoroughly evaluate these architectures using the CloudSEN12 [5] benchmark, specifically assessing performance improvements on "hard" infrequent classes (Thin Clouds and Cloud Shadows) throughout diverse biomes.

   Zantedeschi et al [58] provide us with a comprehensive benchmarking dataset for classifying cloud types that one may use to understand the different clouds shapes and sizes. This dataset consists of 1 year of 1km resolution MODIS hyperspectral imagery which was merged with CloudSat cloud labels. With greater availability of space launch capabilities, satellites being launched in the present carry a host of sensors and camera technology onboard with them to study the Earth. Satellites like the Sentinel-1 launched by the European Space Agency carry with them a single C-band Synthetic Aperture Radar (SAR) which can scan the earth surface in daylight or night settings. The SAR can pierce through clouds and rainfall conditions. Where as a later launched satellite Sentinel-2 has the capability to capture images of various spectra such as infrared, near infrared and visual spectrum. The wavelengths that are measured by Sentinel-2 range from 443.9 nm to detect aerosol particles to 1375.5 nm to detect Cirrus clouds. On top of these other sensors such as Cloud Displacement Index values, 360 degree Azimuth values help researchers with various use cases such as cloud segmentation, land cover land usage classification among others. This thesis is focused on demanding task of developing a robust cloud and cloud shadow segmentation that is capable of functioning in cases of satellite sensor failure. Statistical cloud detection algorithms like FMask 4.0[46] and MAJA[17] are increasing getting overshadowed by deep neural network models such as Resnet[19] and U-net [47]. These models were originally made for image classification tasks but they can be fine-tuned or adapted for cloud and cloud shadow detection. Most major cloud detection algorithms today rely on optical satellite information that is often not enough to detect thin clouds or differentiate between cloud and snow. Li et al [29] developed a sophisticated cloud and cloud shadow detection algorithm based on a combination of UNet [47] and Residual blocks [19]. They employ a feature-level fusion [40] strategy but only use Sentinel-2 optical information as input. Therefore, their model fails in polar climates and cannot accurately distinguish between snow and clouds. Alistair Francis [16] achieves SOTA performance on their cloud and cloud shadow detection algorithm by leveraging multimodal input data, i.e., Sentinel-1, Sentinel-2, and Digital Elevation Maps (DEM).

The primary contributions of this thesis are as follows:

(a) Propose U-Net[47] based robust multi-view fusion model for cloud and cloud shadow segmentation. These models will be tested using two fusion functions, namely Concatenation and Gated Fusion [4].

(b) Propose an Inverse U-Net-based model that tries to learn geometric features like SAR and DEM, along with cloud segmentation and land cover classification.

(c) Used a state-of-the-art global cloud and cloud shadow dataset, CloudSen12[5], to train and test the models.

(d) Investigated the proposed model's performance under a variety of conditions, such as Climate Zones, Cloud Coverages, Difficulty Scores, and Land Cover types.

(e) Tested the model's robustness under a simulated band dropout scenario where Sentinel-2 band information is incrementally removed, and the model's performance is tested on the remaining "active" bands.

The following sections of this thesis document are organised as follows: Section 2 presents the related work summary in the field cloud detection and multi-view image segmentation models, Section 3 is a detailed explanation of the dataset being used in the thesis and the proposed model architecture, Section 4 consists of the training procedure and obtained results, and a final conclusion is given in Section 5.

Abbreviations

ML	Machine Learning
RS	Remote Sensing
S1	Sentinel 1
S2	Sentinel 2
SAR	Synthetic Aperture Radar
DEM	Digital Elevation Maps
AMG	All Modality Gated
AMC	All Modality Concat
RSFM	Remote Sensing Foundation Model

Chapter 2
Related Work

This section presents an overview of current deep learning techniques and state-of-the-art cloud and cloud shadow segmentation approaches. Further, a closer look at data fusion techniques and their implementation in the cloud and cloud-shadow segmentation domain is provided.

2.1 Satellite Imagery

The Earth is being regularly monitored and photographed from space using artificial satellites. There are about 12,149 satellites circling Earth in various orbits, namely Low Earth Orbit, Medium Earth Orbit, and Geostationary Orbit. Each satellite is equipped with a sensor on board that collects enormous amounts of data daily. These sensors can be radar or optical cameras. The raw data captured by these satellites is transmitted back to Earth-based ground stations via radio signals. Satellites either temporarily store the information on board until they are requested by the ground station or transmit it instantly. This raw data is in binary format (Level 0) and is processed to remove errors caused by atmospheric conditions, sensor noise, and geometric distortions. The final image ready for analysis is Level 1C, in array form. There can be more than one array for every image, depending on the type of sensor aboard the satellite. The values in the array range from 0 to 255, where 0 represents no radiation (black), and 255 represents the highest radiation (white). These arrays also include metadata, such as latitude and longitude coordinates assigned to each pixel. There are various types of satellite imagery available to us. Some of the most important ones are discussed below.

2.1.1 Optical Imagery

These images are captured by satellites that detect light reflected from Earth's surface, just as our eyes do. They are utilised for a plethora of tasks such as urban planning, weather forecasting, forestry, and agriculture, just to name a few.

Multi-spectral Satellites: Modern optical satellites such as Sentinel-2, WorldView and Landsat series are multispectral in nature, i.e. they consist of sensors that are capable of detecting a variety of wavelengths along with the traditional red, green and blue bands. Sentinel 2 consists of 13 spectral bands with varying wavelengths and purposes. Table 2.1 describes the bands analysed by Sentinel-2.

Band	Resolution	Central Wavelength	Description
B1	60 m	443 nm	Ultra Blue (Coastal and Aerosol)
B2	10 m	490 nm	Blue
B3	10 m	560 nm	Green
B4	10 m	665 nm	Red
B5	20 m	705 nm	Visible and Near Infrared (VNIR)
B6	20 m	740 nm	Visible and Near Infrared (VNIR)
B7	20 m	783 nm	Visible and Near Infrared (VNIR)
B8	10 m	842 nm	Visible and Near Infrared (VNIR)
B8a	20 m	865 nm	Visible and Near Infrared (VNIR)
B9	60 m	940 nm	Short Wave Infrared (SWIR)
B10	60 m	1375 nm	Short Wave Infrared (SWIR)
B11	20 m	1610 nm	Short Wave Infrared (SWIR)
B12	20 m	2190 nm	Short Wave Infrared (SWIR)

Table 2.1: Sentinel-2 optical satellite spectral bands and corresponding information

The combination of these bands emphasises distinct features, thereby highlighting characteristics normally undetectable to the human eye. An example for such a combination would be Normalised Difference Vegetation Index (NDVI) as shown in equation 2.1. It is formed by combining red and near-infrared (NIR) bands to highlight vegetation in our region of interest. This is because vegetation reflects near-infrared light and absorbs red light. As a result, one can effectively track deforestation and the health of trees in a specific area. Another metric is the Modified Normalised Difference Water Index (MNDWI), represented in equation 2.2, which enhances water features while simultaneously suppressing built-up areas and forests. MNDWI uses green and SWIR bands because water reflects green light the most and absorbs SWIR light. At the same time, land features strongly reflect SWIR light. Sentinel-2 information can either be an L1C product or an L2A product. The Sentinel-2 L1C product retains atmospheric influences such as Water Vapour and Aerosols. For this thesis, we have used the Sentinel-2 L1C product, as it provides Top-of-Atmosphere reflectance (TOA) and retains atmospheric information. L2A products discard this valuable information and are therefore not used for cloud and cloud shadow detection.

$$\mathrm{NDVI} = \frac{\mathrm{NIR} - \mathrm{Red}}{\mathrm{NIR} + \mathrm{Red}} \tag{2.1}$$

$$\mathrm{MNDWI} = \frac{\mathrm{Green} - \mathrm{SWIR}}{\mathrm{Green} + \mathrm{SWIR}} \tag{2.2}$$

Hyperspectral Satellites

Hyperspectral satellites measure long, continuous spectral bands, whereas multispectral satellites measure only a set of specific spectral bands or wavelengths. These satellites typically capture more than 100 bands for every Area of Interest

(AOI). Each band captured is very narrow, aka only 5-10 nanometers wide. This high-precision image capture yields a very high spectral resolution and a continuous spectral signature for each pixel. But they also result in exponentially more data that is stored and transmitted back to ground stations. These satellites are typically used for high-precision tasks such as mineral exploration, water quality, and environmental mapping. Some examples of these satellites are HySIS, Firefly and EnMAP. Figure 2.1 shows some examples of hyperspectral images that were captured over the river Ganges in India by the Firefly satellite. One can observe how different aspects of the riverbed are exposed by the different bands of the hyperspectral image.

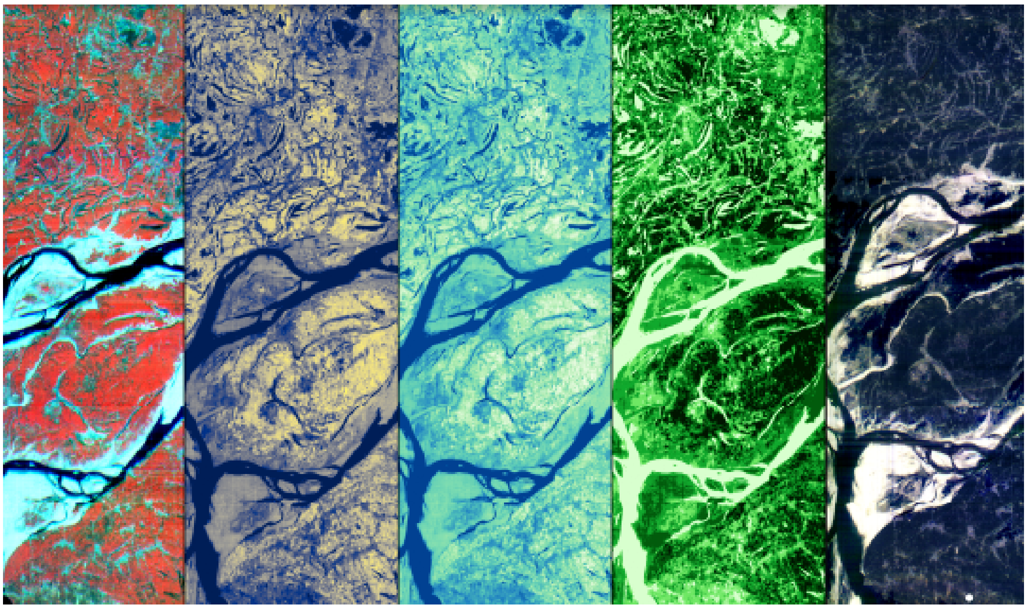

Figure 2.1: Hyperspectral images captured by Pixxel Space's Firefly series of hyperspectral satellites [1]

2.1.2 Radar Imagery

Radar imagery differs from optical imagery because it uses an active sensor rather than a passive one to scan the Earth. These satellites send out radio waves that bounce off the Earth's surface and return to the satellite as echoes. They are not limited by sunlight; i.e., they work in both day and night conditions and can see through clouds. This helps them detect subtle changes occurring on the Earth's surface that might not be visible to Optical satellites. Examples of such satellites include the Copernicus Sentinel-1 constellation and NISAR. Both these satellites are equipped with Synthetic Aperture Radar (SAR) on board. SAR uses radio or microwave pulses to observe the Earth. On the other hand, Unmanned Aerial Vehicles (UAVs) are often equipped with a sensor called Light Detection and Ranging (LiDAR). LiDAR sensors emit Infrared pulses to examine the Earth. LiDAR sensors are also commonly used in self-driving vehicles to generate depth maps of their surroundings.

2.2 Computer Vision

Computer vision is the discipline of computer science that deals with the acquisition, processing, and analysis of images. The primary goal in this regard is to teach computers how humans perceive the visual world and generate valuable insights [3]. Computer vision greatly benefits the world, as it is utilised in a variety of tasks, such as autonomous driving, anomaly detection, and 3D mapping. The most common established tasks of computer vision are as follows:

Image Classification: This task refers to the objective of a machine to observe an image and accurately predict the class it belongs to. Simply put, image classification assigns a label to an entire image. An example of image classification is shown in Figure 2.2.

Figure 2.2: Satellite Image Classification [56]

Object Detection: This task builds upon image classification and is used to detect specific classes within an image. These detected objects are framed in boxes, which are known as bounding boxes. One example in remote sensing is the detection of solar panels on the roofs of residential buildings, as shown in Figure 2.3.

Semantic Segmentation: This task analyses every pixel in a given image and assigns it to a specific class. Our intended task of cloud and cloud shadow segmentation is an accurate example of semantic segmentation, where each pixel is labelled as a thick cloud, a thin cloud, a cloud shadow, or background. Figure 2.4 illustrates an example of semantic segmentation using drone video footage vividly.

2.3 Machine Learning in Computer Vision

The advent of AlexNet [26] in 2012 opened the door to the use of neural networks in image processing and computer vision tasks. In the following section, I give a brief overview of the functioning and eventual evolution of neural networks to handle complex tasks and give remarkable insights.

Figure 2.3: Rooftop Solar Object Detection with Bounding Boxes [45]

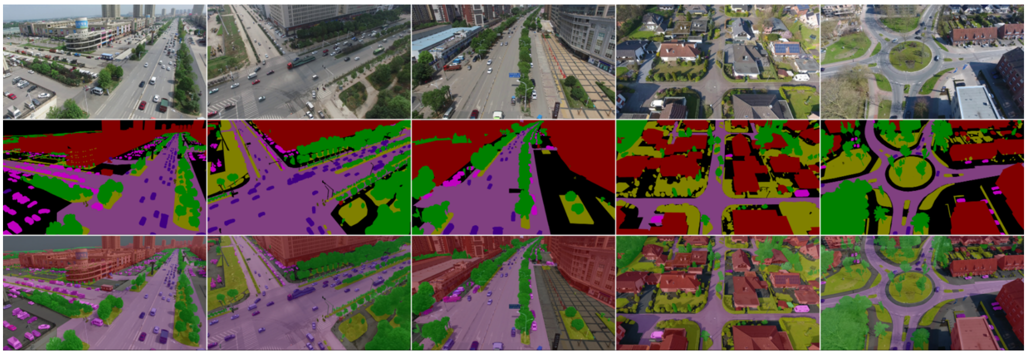

Figure 2.4: Semantic Segmentation of Urban Drone Footage [36]

2.3.1 Learning visual representations with neural networks

Artificial neural networks(ANNs) are named such because they are mathematical models that are inspired by the human brain and its constituent nerve cells or neurons. The major advantage of using artificial neural networks(ANNs) is their ability to learn representations from data without using feature extractors that are explicitly designed for a particular use case. ANNs are able to learn how to extract important features from raw data hierarchically. It is able to accomplish such a feat by using multiple interconnected processing layers. Every layer transforms the incoming data and learns an abstract representation from the data. Every constituent neuron is linked to all upstream and downstream neurons in the neural network. Therefore, an artificial neural network(ANN) can be divided into three broad regions, aka the "input layer", "hidden layer", and "output layer". An example of a typical neural network is shown in Figure 2.5.

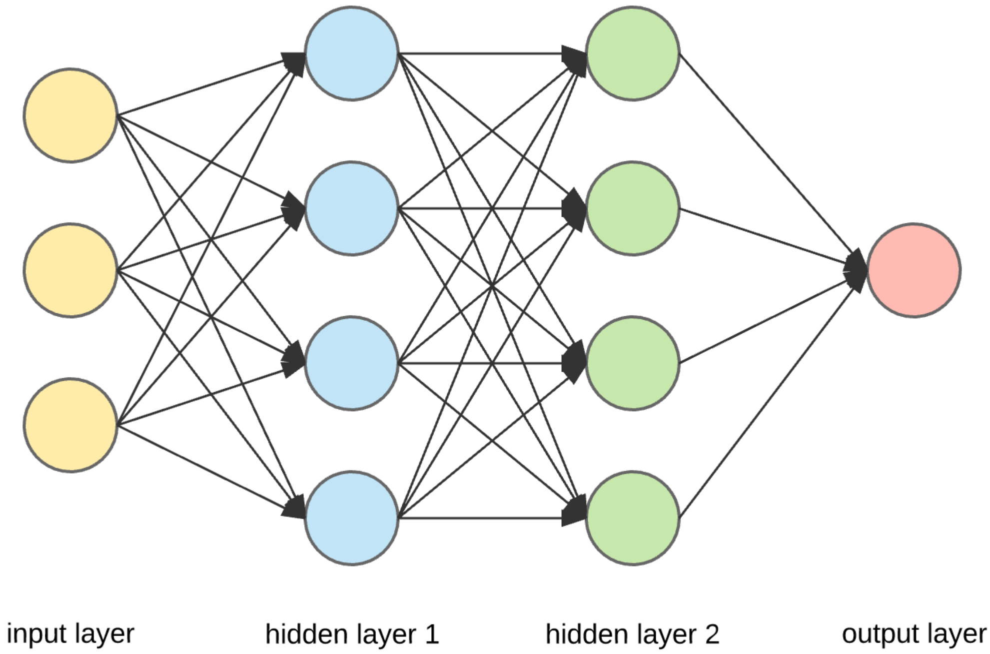

Figure 2.5: Structure of a Neural Network

ANNs learn the relations between input and output data using a technique known as backpropagation[48]. Every connection between two neurons is balanced by "weights", which are adjusted during training in order to produce better representations and develop a more precise mapping of the data. The error between the network's output and the actual ground truth value is corrected using a loss function. This is how an ANN can inherently represent any mathematical function [?]. In the case of images, ANNs learn their representations by first extracting a set of lower-level features such as edges, which subsequently make up parts and ultimately objects inside the image. A visual representation of how an ANN learns to extract representations of an image is shown in Figure 2.6.

2.3.2 Convolutional neural networks

Convolutional neural networks (CNNs) are a specialised type of deep neural network that works very well for processing images and videos. The major difference between a CNN and a fully connected neural network is the use of convolutional operations in a convolutional layer. The fundamental parts that compose a CNN are as follows:

Convolution Layer: A 3-dimensional tensor or "kernel" consisting of a set of weights is matrix multiplied over the input image tensor to generate a scalar product. This tensor is then moved along the image tensor with a pre-defined stride until a "feature map" is obtained. The output tensor is formed by stacking the results of multiple kernels. This method of sharing weights with different

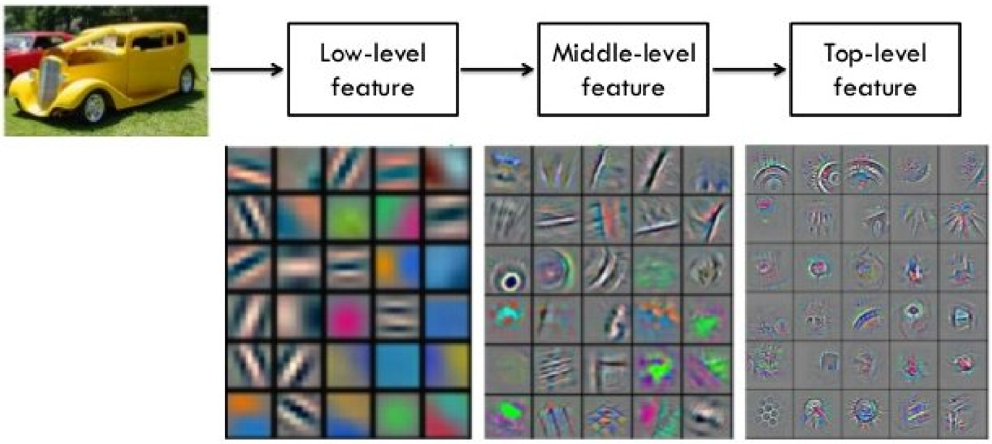

Figure 2.6: Feature extraction from an Image by an ANN [59]

parts of the input tensor enables the detection of local features while reducing the number of trainable parameters. Further, an element of non-linearity is obtained when the output tensor is multiplied element-wise by an activation function. The most commonly used activation function is the Rectified Linear Unit (ReLU) [42], defined in equation 2.3 and illustrated in figure 2.7.

$$f(x) = \begin{cases} 0 & \text{if } x < 0 \\ x & \text{if } x \geq 0 \end{cases} \tag{2.3}$$

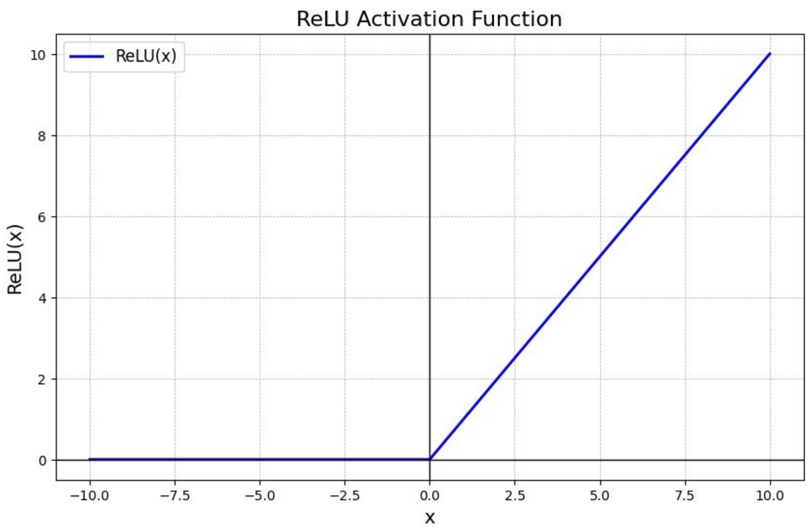

Figure 2.7: Rectified Linear Unit (ReLU) Activation Function

Pooling Layer: A pooling layer is typically placed immediately after a convolutional layer, and its purpose is to decrease the number of trainable parameters and make it much more manageable. It operates independently on each activation map and merges semantically similar features. This reduces the overall computational cost of the CNN. The most commonly used pooling function is named "Max Pooling", demonstrated in Figure 2.8.

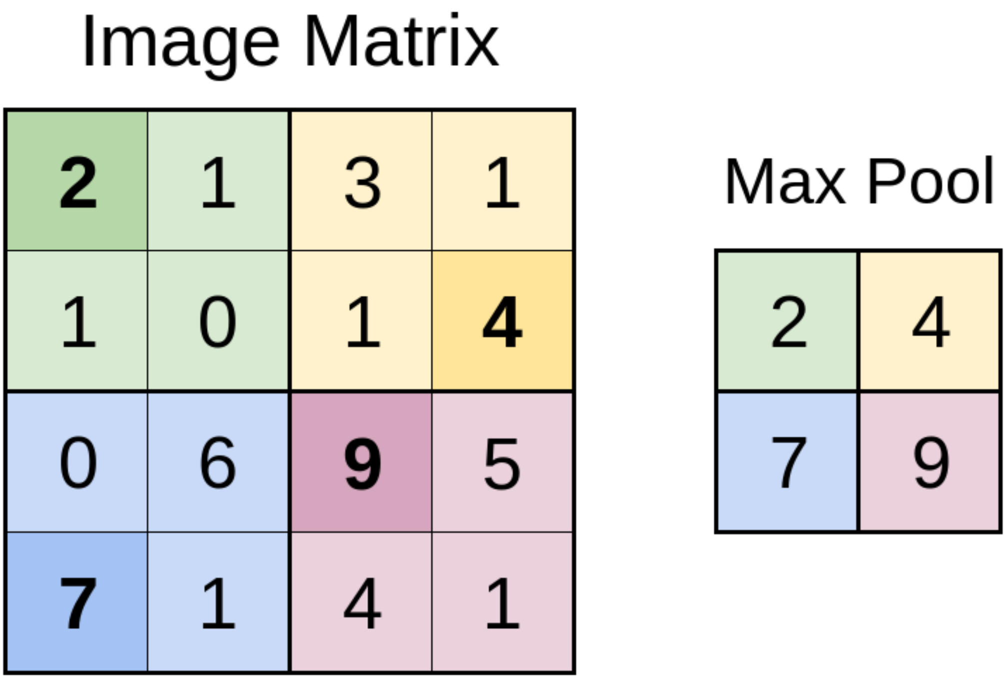

Figure 2.8: Representation of Max Pooling Operation

Dropout: This layer has a specific use case in neural networks, i.e. to prevent overfitting. Overfitting occurs when the neural network learns to perfectly imitate the training data and fails to generalise to new, unseen data. A dropout layer's objective is to force the ANN to forget certain features and help it generalise to the training data. Typically, a dropout layer "switches off" a certain number of neurons in an ANN, i.e., sets their weights to 0. This results in a model that also accurately predicts on new, unseen data.

2.3.3 Semantic Segmentation with CNNs

Before the popularity of Artificial Neural Networks (ANNs), semantic segmentation tasks were performed using statistical algorithms such as K-means clustering [41] and Support Vector Machines (SVMs) [20]. One major drawback of ANNs such as AlexNet [26] and VGG [51] was that they used fully connected layers at the end of their models. This network architecture was not sufficiently effective at achieving accurate semantic segmentation. Long et al [33] demonstrated that replacing the fully connected layers with convolutional layers made the networks accept images of all possible sizes and also output a segmented image of the same size. They also introduced the concept of "skip-connections". Subsequently, a breakthrough was achieved in 2015, when Ronneberger et al. [47] published the U-Net paper on biomedical image segmentation. This revolutionised the field of semantic segmentation, becoming the bedrock for **Encoder-Decoder** architectures still in use today. U-Net [47], as the name suggests, consists of a U-shaped network where the left-hand side, aka the Encoder, is used to capture contextual information about the input image. Whereas the right-hand side, aka the decoder, restores the image back to its original dimensions while simultaneously preserving its global and local features. This symmetrical U-shaped structure, combined with skip connections at every layer of the network, helped it preserve both local and global contextual information, making it the most widely used architecture for image segmentation. U-Net's architecture can be further visualised from Figure 2.9 given below. Here, every multi-channel feature map is represented as a blue box. The image size at each layer is shown at the lower-left corner of the boxes. The feature maps are copied with the help of skip connections, which are represented in white. The numbers at the top of the boxes represent the number of channels. Finally, the arrows indicate the different operations occurring at each layer.

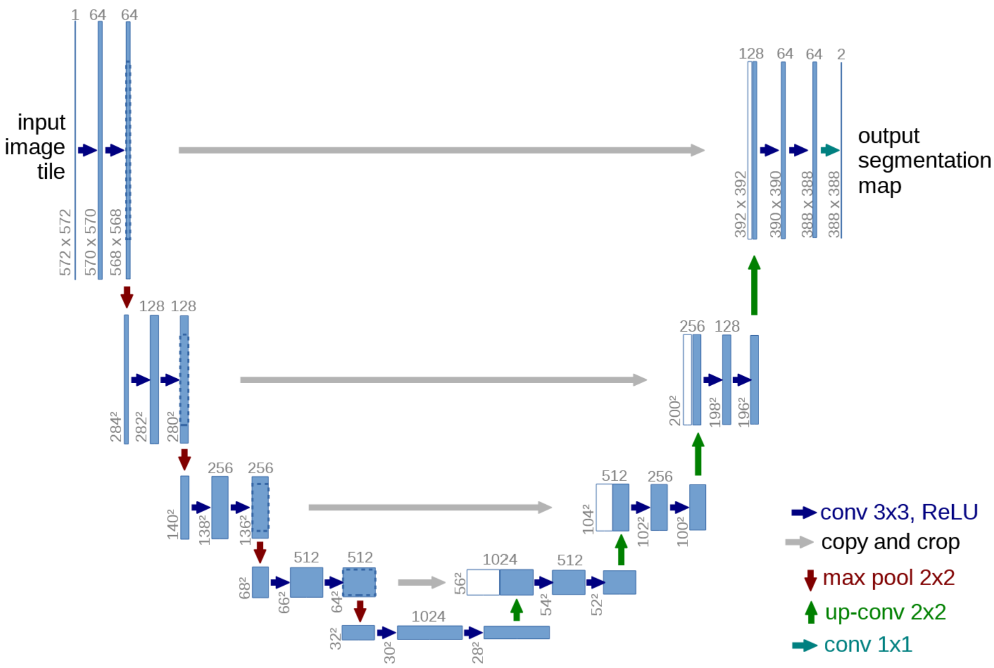

Figure 2.9: A typical Unet architecture with skip connections [47]

While the U-Net and its variants have become the industrial standard for semantic segmentation tasks, they are traditionally designed for RGB (Red, Green, Blue) images. In the domain of remote sensing and earth observation, satellites are equipped with multiple bands or channels, as mentioned in the sections above. Further, relying on a single data source leads to poor results due to factors such as cloud cover. Therefore, to make the most out of the multi-modal nature of satellite information, one must extend these architectures beyond simple inputs. This leads us to the concept of Data Fusion or Multiview Learning.

2.4 Data Fusion or Multi-view Learning

Data fusion, or Multi-view Learning, refers to the process of integrating multiple sources of data (aka multiple views) to obtain more consistent and accurate results for a given problem statement, rather than using just one data source.

In the case of cloud and cloud shadow segmentation, data fusion refers to the process of combining different instruments, such as Synthetic Aperture Radar (SAR) and LiDAR, with optical satellite imagery to achieve better results. But data fusion covers a vast area that is under active research to this day.

2.4.1 Data Fusion Challenges

There are various intrinsic challenges to data fusion or multi-view learning. The data received from satellites could have different spatial or temporal resolutions. This remote sensing data cannot be stacked on top of each other and fused easily. This makes the researchers construct specialised models for each type of data source, such as a CNN for spatial data combined with a simpler MLP (Multi-Layer Perceptron), aka a fully connected neural network, for additional tabular data, and finally a RNN (Recurrent Neural Network) for the temporal data. Even if the data is only varied in the spatial domain, they could vary information levels, i.e. high-level features vs low-level features. One might need a deeper CNN for high-resolution optical data, whereas a shallow CNN is needed for low-resolution water mask. As the models become more complex with additions of new views, there is a higher risk of over-fitting. Rather than being a model that is relatively generalised when trained on a single view, multi-view models tend to overfit if there is not enough labelled data. Wang et all [52] shows us how each view generalises at different levels, and one can overfit before the other. An example that they use is an audio network that can overfit by memorising the data quickly, whereas the video network is still learning useful patterns for generalisation. Therefore, this complex task of multi-view learning requires researchers to correctly figure out where and how to fuse the different views.

2.4.2 Fusion Locations aka Where to Fuse

The most obvious question that emerges in multi-view learning is where to fuse the different views or data. The options that are available to us are either early fusion, fusion in the middle layers or in the late stages. These correspond directly to input level fusion, feature level fusion and decision level fusion, respectively.

Input Level Fusion or early fusion is the most naive approach out of the three available options. This process involves stacking all the different views available to us and then pushing all of them to a single model directly. This, in essence, turns the task of multi-view learning into a single-view learning. On one hand, input level fusion makes it very difficult for the model to learn explicit features from all the different views available to it, but it also reduces the complexity of the model and therefore its implementation becomes easier. The model does not explicitly learn fusion and therefore requires increased complexity to understand the relationship between the different views. In remote sensing, the most important thing to keep in mind is that the different views must be temporally and spatially aligned. Without this alignment, the model is most likely to fail to learn any meaningful information. This has been illustrated in figure 2.10.

Feature Level Fusion refers to the type of fusion where every view has its own representation. In deep learning terms, every view has its own encoders or backbone models, which learn the features of that individual thoroughly. These feature extractor modules are then fused or combined using various techniques like concatenation or intelligent fusion and then passed on towards a predictive model/decoder. This approach is less common than input-level fusion. This results in models of varying complexity and the ability to share information between different views as represented in Figure 2.10.

Decision Level Fusion or late stage fusion refers to the method of data fusion where individual views have their own single-view end-to-end models. The output of these single-view models is combined just before the prediction head, and a singular loss is calculated on the stacked output. This is computationally the most expensive type of fusion as researchers have to run n parallel single-view models for n number of views as input. This is also the least preferred method of data fusion.

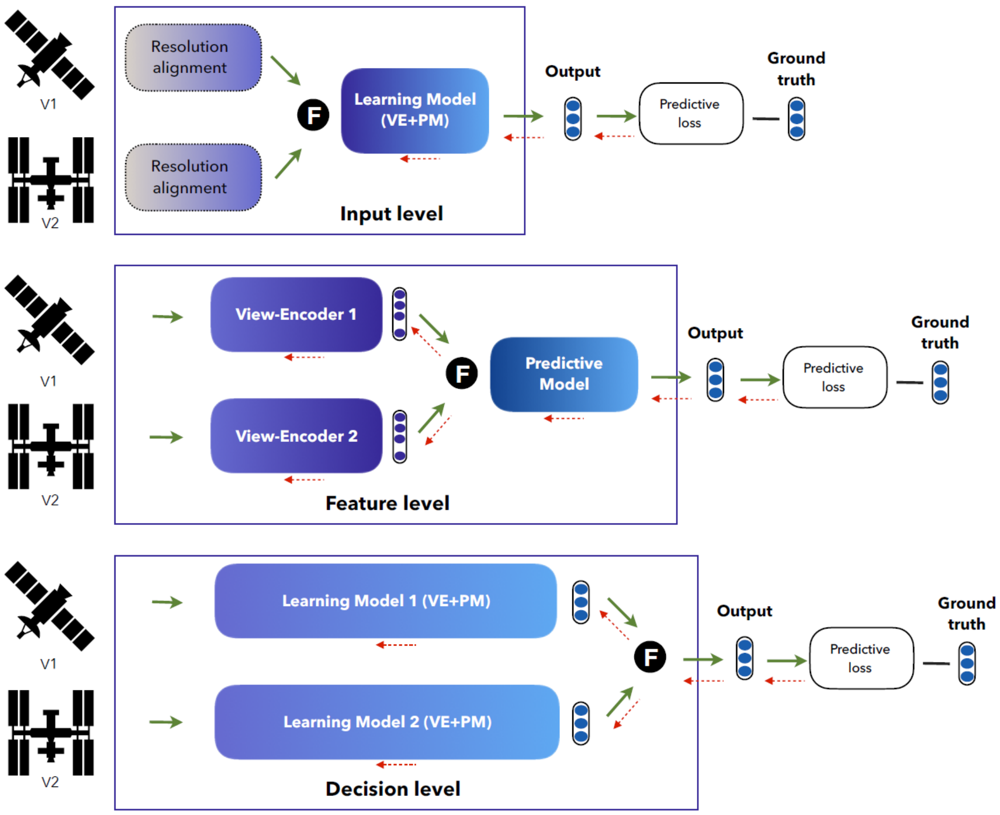

Figure 2.10: Illustration describing the different data fusion techniques as referred from Francisco et al [39]

2.4.3 Fusion Techniques aka How To Fuse

Once the location of the data fusion is determined, one must also determine the method to be used to fuse/merge the different views or modalities. These fusion techniques range from naive solutions such as concatenation, element-wise product, and uniform sum, to intelligent ones such as weighted sum, attention-based fusion, and gated fusion [4].

2.5 Literature Review

2.5.1 Cloud Detection and Cloud Cover Segmentation

Historically, cloud detection in satellite imagery primarily utilised statistical algorithms. Notable examples include FMask [46] and Sen2Cor [37], which were widely adopted for cloud cover identification before the introduction of deep learning. The development of Convolutional Neural Networks (CNNs), especially architectures like ResNet [19], has fundamentally transformed the field, establishing these models as the foundation of contemporary detection algorithms.

Expanding on this advancement, Rozenhaimer et al. [49] introduced a domain adaptation CNN approach that utilises WorldView-2 (WV-2) and Sentinel-2 (S-2) data. By integrating both spectral and spatial information, the algorithm learns deep invariant features, enabling effective domain adaptation. This method offers a significant advantage: accurate prediction across multiple satellite platforms without requiring separate training, resulting in improved accuracy for both clear and cloudy pixels compared to single-source training. In a similar vein, Chai et al. [8] developed a CNN model that leverages multi-level spatial and spectral features from all available bands, facilitating granular classification by labelling pixels into four categories: cloud, thin cloud, cloud shadow, or clear sky.

Advanced Architectures: Attention Mechanisms and Efficiency: To overcome the limitations of traditional methods, particularly in edge detection and small cloud identification, recent studies have incorporated attention mechanisms. Zhang et al. [60] introduced a refinement segmentation network based on ResNet-18, integrating a Multi-scale Global Attention Module to enhance channel and spatial information, as well as a Strip Pyramid Channel Attention Module. This combination has led to notable improvements in detecting small clouds and delineating fine edges. Parallel research has emphasised structural efficiency. Dev et al. [10] presented CloudSegNet, a lightweight architecture optimised for efficient segmentation. This framework demonstrates versatility by outperforming existing methods on both daytime and nighttime imagery within a single unified system.

Semantic Segmentation and Shadow Detection: Encoder-decoder networks, exemplified by U-Net [47], have significantly enhanced segmentation accuracy. Building on this foundation, Ouyang and Li [43] proposed the DSSN-GCN framework, which integrates a Deep Semantic Segmentation Network (DSSN) with a Graph Convolutional Neural Network (GCN). They specifically introduced the Attention Residual U-shaped Network (AttResUNet) to extract high-level features and initialise graph nodes, thereby effectively combining appearance extraction with topological relationship modelling. Cloud shadow detection, which frequently encounters challenges due to insufficient feature extraction, has also received focused attention. Luo et al. [35] introduced the Deeply Supervised CNN for Shadow

Detection (DSSDNet) tailored for aerial images. Utilising an encoder-decoder residual (EDR) structure and a deeply supervised progressive fusion (DSPF) process, this model achieves more consistent shadow detection than current state-of-the-art methods.

2.5.2 Data Fusion and Cloud Removal

Data fusion integrates multiple data sources, including Synthetic Aperture Radar (SAR), LiDAR, RGB, and infrared images, to produce results that surpass those of single-source analysis. Li et al. [28] offer a comprehensive review of deep learning techniques for data fusion tasks, encompassing spatiotemporal analysis, pansharpening, and object extraction. One of the primary applications of data fusion is cloud removal, achieved by combining SAR data, which can penetrate clouds, with optical imagery. Although Sentinel Hub provides the s2cloudless algorithm, recent academic research has aimed to advance beyond standard removal techniques.

Xu et al. [54] introduced a global-local fusion algorithm, GLF-CR, which utilises the SEN12MS-CR dataset [14], a companion to CloudSEN12 [5]. Their approach incorporates a Transformer-inspired SAR-guided Global Context Interaction (SGCI) block to maintain structural consistency between cloud-removed regions and the original optical images. Additionally, to address the inherent speckle noise in SAR data, they developed a SAR-guided Local Feature Compensation (SLFC) block that generates reliable texture data while reducing noise artefacts. To improve detection in challenging terrain, Li et al. [30] proposed the Multi-Scale Convolutional Feature Fusion (MSCFF) method. This deep learning approach uses a symmetric encoder-decoder module to iteratively minimise loss. It has proven particularly effective for remote sensing imagery from various sensors, achieving higher accuracy than rule-based methods, especially in areas with bright surface cover that often confuses traditional algorithms.

The best models, their methods, the problems they addressed, and the key advantages they describe are listed below in Table 2.5.2.

Reference	Method / Model	Problem Addressed / Focus	Key Advantages
Rozenhaimer et al. [50]	Domain Adaptation CNN	Lack of generalisation between different satellite sensors (WV-2, S-2).	Allows adaptation without separate training; improves accuracy for both clear and cloudy pixels.
Chai et al. [8]	Multi-level CNN	Need for granular classification beyond binary masks.	Uses global/spectral features to distinguish 4 classes (cloud, thin cloud, shadow, clear).
Zhang et al. [60]	ResNet-18 + Attention Modules	Inability of traditional methods to detect small clouds and fine edges.	**Multi-scale Global Attention** improves spatial info; **Strip Pyramid** module detects small clouds effectively.
Dev et al. [10]	CloudSegNet	Computational inefficiency and varying lighting conditions.	Lightweight architecture; effective for both daytime and night-time imagery.
Ouyang & Li [43]	DSSN-GCN (AttResUNet)	Modelling complex topological relationships in segmentation.	Combines appearance extraction (DSSN) with topological modelling (GCN) for better semantic segmentation.
Luo et al. [35]	DSSDNet (EDR + DSPF)	Insufficient feature extraction specifically for **shadows**.	Deep supervision and progressive fusion yield more consistent shadow detection.
Xu et al. [54]	GLF-CR (SAR + Optical)	Recovering information under clouds; SAR speckle noise.	**SGCI** block ensures structural consistency; **SLFC** block reduces SAR speckle noise.
Li et al. [28]	MSCFF	Accuracy issues over bright surfaces (e.g., snow/sand).	Symmetric encoder-decoder outperforms rule-based methods in bright surface areas.

Table 2.2: Summary of Reviewed Methodologies

Chapter 3
Methodology

As discussed in the previous section, multi-view deep learning has significantly advanced cloud and cloud shadow segmentation. However, the existing literature identifies two critical limitations: first, most algorithms based on Convolutional Neural Networks (CNNs) or transformers are trained solely on optical data. These algorithms struggle with spectral uncertainty in complex environments such as snow or bright urban areas. This leads to inaccurate segmentation results. Secondly, the multi-view models are mostly trained on Optical and Synthetic Aperture Radar (SAR). Other modalities, such as Digital Elevation Maps (DEMs) and secondary information, such as land cover, are rarely used. Thirdly, these multi-view models are rarely tested for robustness to information loss.

This thesis seeks to address these gaps by combining multiple views beyond SAR and Optical. All these views are combined and tested using two fusion techniques: a naive concatenation-based fusion and an intelligent Gated Fusion [4]. Gated Fusion is theoretically better, as it intelligently prioritises and assigns greater weight to views that contribute more to the final segmentation results. Furthermore, the robustness of these two models to information loss is tested by removing one optical band at a time and checking their performance. Lastly, we investigate the potential of Multi-task Learning in this domain by inverting this segmentation architecture to predict geometric features such as SAR, DEM, and Landcover using optical information as input.

This chapter provides an overview of the experimental framework used to validate these contributions. Section 3.1 describes the CloudSen12 dataset [5] used to test our models and hypotheses. Whereas Section 3.2 details three model architectures designed to test the multi-view learning task, which is robust to information loss, and the pretext learning task, respectively.

3.1 Dataset

For this thesis, we use the Cloudsen12 dataset [5], a global dataset for cloud and cloud-shadow segmentation. It consists of Sentinel-1 Synthetic Aperture Radar

(SAR) information, Sentinel-2 optical information distributed across 13 bands, along with DEM (Digital Elevation Maps) and other derivative information such as Land Cover [57], Azimuth, and Water Occurrence [44]. As discussed in the previous sections, Sentinel-2 is an optical satellite that consists of 13 bands, ranging from the visible spectrum (Red, Green, Blue) to Near Infrared (NIR) and Short Wave Infrared (SWIR). Sentinel-1 Synthetic Aperture Radar data is divided into 3 bands: Vertical Transmit/Vertical Receive (VV) Polarisation, Vertical Transmit/Horizontal Receive (VH) Polarisation, and Angle of Incidence. Elevation map data is stored in meters and obtained from the MERIT Hydro datasets [55]. Land Cover information is obtained from the ESA WorldCover [57] 10m resolution product. Azimuth values range between $0°$ to $360°$, and Water Occurrence refers to water presence frequency obtained from JRC Global Surface Water [44]. A detailed overview of the different modalities is provided in the following chapter (Figure 4.3). This global dataset consists of images from 1827 unique Regions of Interest (ROIs) in the training set and 173 unique ROIs in the test set, spanning the world. Each ROI was selected specifically to have a diverse set of locations and natural features. Further, for every ROI, there are 5 types of cloud coverage available in the dataset, i.e. cloud-free (0%), almost-clear (0–25%), low-cloudy (25–45%), mid-cloudy (45–65%), and cloudy (>65%). Here, the percentage refers to the portion of a sample image obstructed by clouds as shown in Figure 3.1.

The authors further provide these labels in three forms: high-quality annotated (with human assistance), scribble-annotated, and no annotation (for unsupervised learning). For this thesis, I have chosen only high-quality, human-annotated labels for Supervised Deep Learning tasks.

3.2 Model Architectures

To rigorously evaluate the efficacy of the multi-view models and the pretext task model, we developed distinct architectures for each. The multi-view models allow us to test the fusion strategy and robustness, and the pretext task model allows us to test the learning capability.

3.2.1 Multi-view Cloud and Cloud Shadow Segmentation Models

The first two models follow a multi-view encoder-decoder structure. They use 1 encoder per view but are restricted to a single decoder because the final output is a segmentation mask with 4 classes. Here sentinel-2 optical information is fused along with Sentinel-1 SAR information, DEM, Water Occurrence, Azimuth and Land cover information. They have 6 encoders and 1 decoder each. Each Encoder and Decoder consists of two sets of convolution operations. After every convolution operation, we used the ReLU (Rectified Linear Unit) [42] activation function and Batch Normalisation [22]. The feature maps were downsampled using Maxpool2D and upsampled using Pytorch's upsampling operation. We also used Dropout after every convolutional layer, with 30% at the top layer, 20% in the middle layers, and 10% at the bottleneck. The major difference between the two models lies in the bottleneck-level fusion strategy. The first fusion strategy is simple concatenation, where all modalities are naively stacked into a single tensor. This significantly increases the model's size, as each view contains hundreds of channels at the bottleneck layer. Subsequent decoder layers also employ concatenation fusion

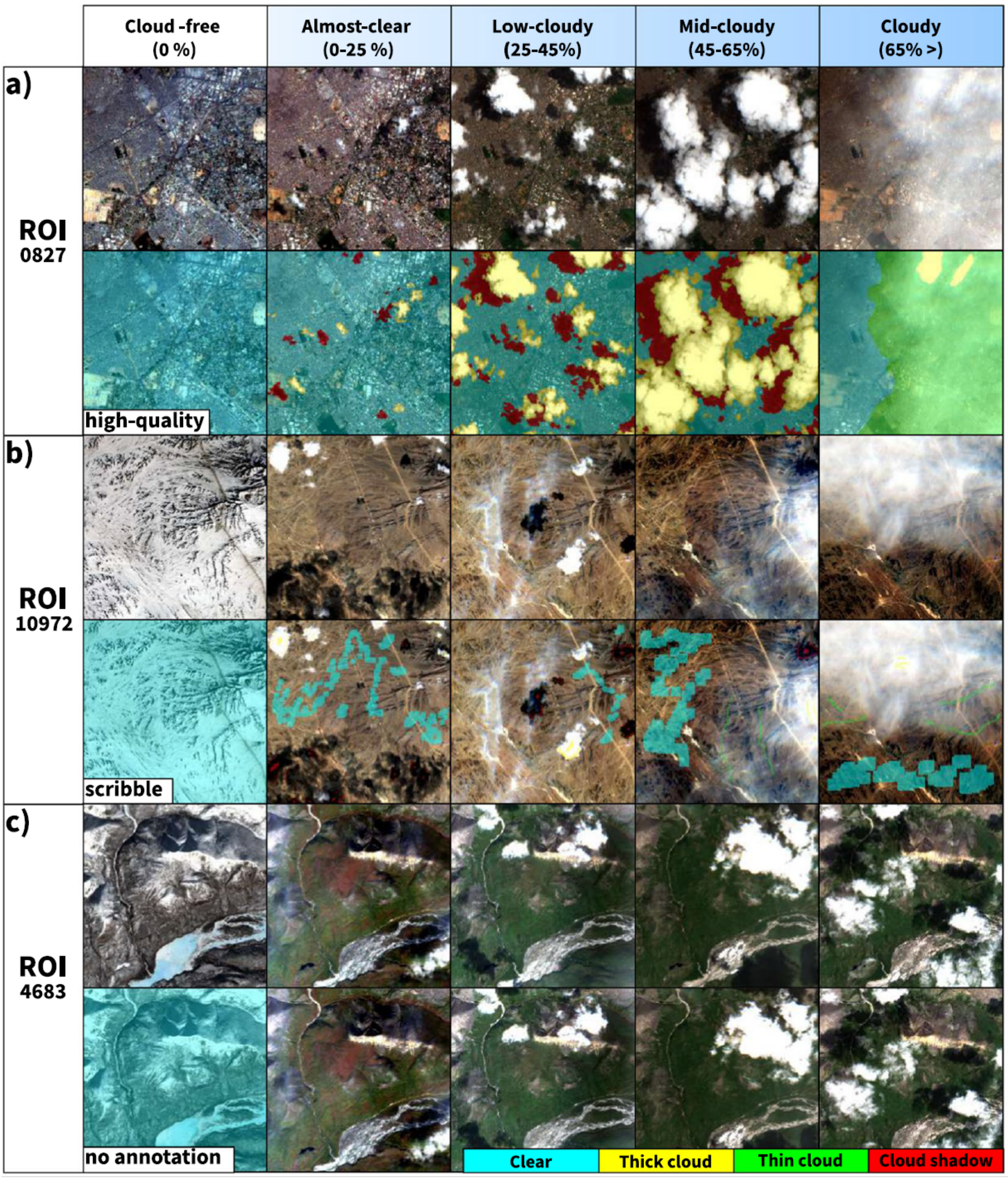

Figure 3.1: Example plot showing the 5 different types of Cloud Coverages along with 3 types of Labelled Images.

at the skip connections. The second fusion strategy is Gated Fusion [4], where each modality is dynamically combined by the model itself. The Gating function is a learnable hyperparameter that assigns greater importance to modalities that contribute to the final output and less importance to those that do not. Figure 3.2 is an illustration of a gating function with 6 inputs as used in this thesis. Figure 3.3, on the other hand, is a detailed illustration of the first set of models. The Fusion operation P can be replaced with either the Concatenation or the Gated Fusion technique.

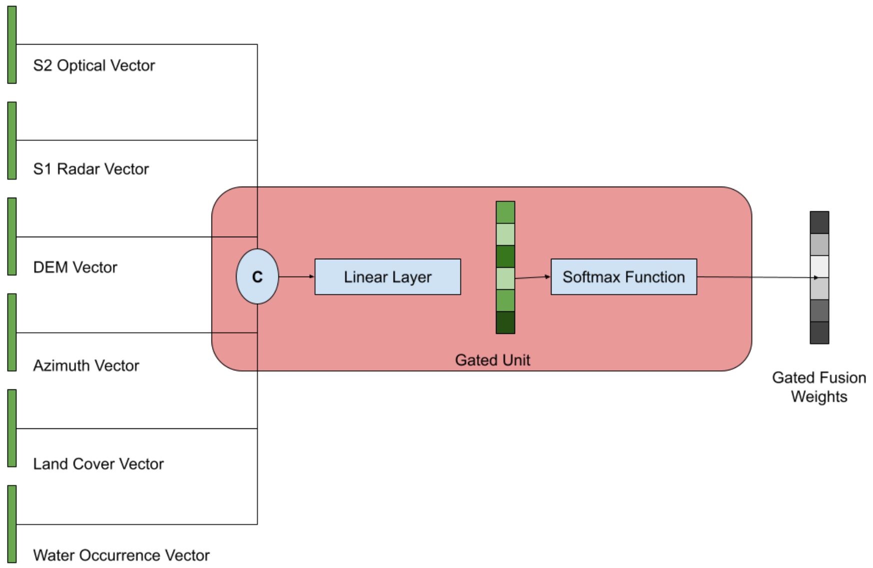

Figure 3.2: **Gated Fusion Module** with six input views. Here, C refers to the Concatenation function.

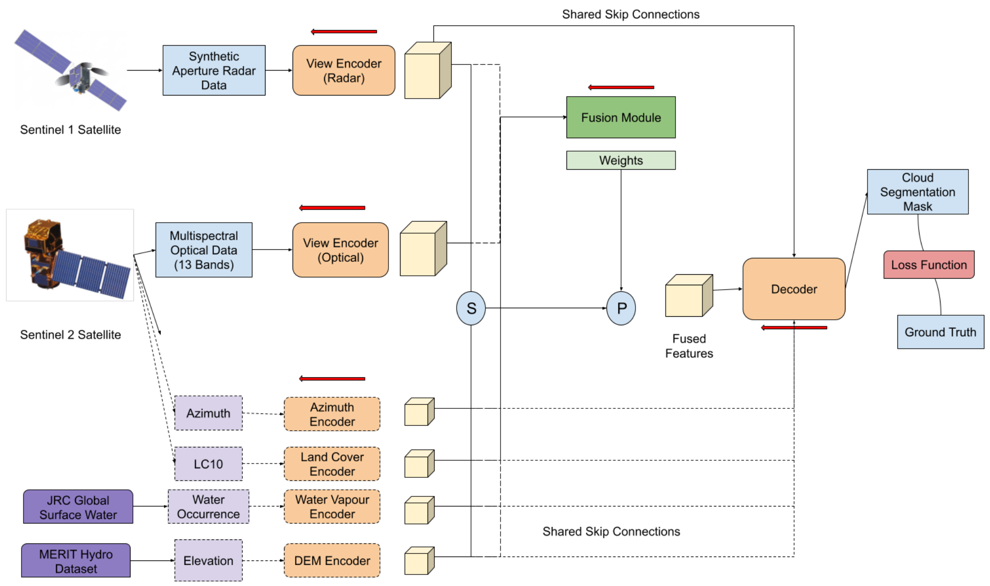

Figure 3.3: **Multi-view Model Diagram** with every modality highlighted in detail. Here, S refers to the Concatenation function, and P refers to the fusion operation.

3.2.2 Pretext Task Model

The third model represents an experimental architecture: an "Inverted U-Net" designed to test if a model trained solely on optical information can learn to

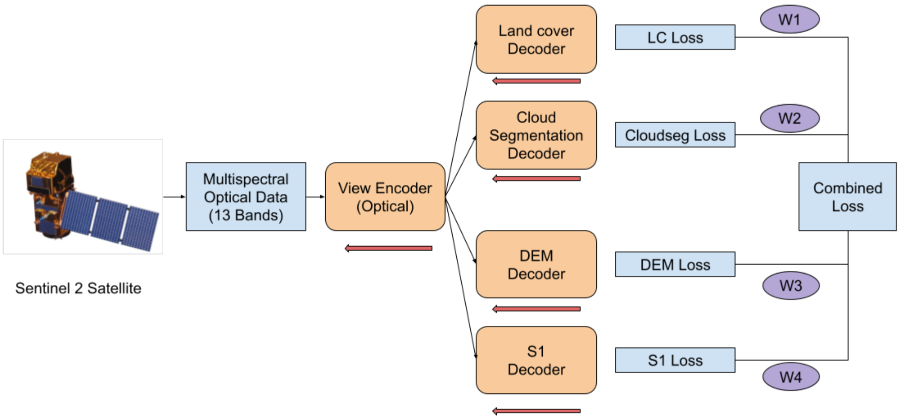

Figure 3.4: **Pretext Task Model Diagram** with weighted combined loss. Here, W1, W2, W3 and W4 refer to the weights assigned to each loss.

reconstruct other views, such as DEMs and SAR, along with derivative information, such as Land Use Land Cover. In this case, the model architecture is reversed: there is only 1 encoder for the Sentinel-2 optical information, with a separate decoder for each of the previously mentioned tasks. The Main Loss is a weighted sum of the individual four losses from each of the four decoders. In our experiment, we assigned the cloud segmentation loss a weight of 1.0, and the other 3 pretext losses a weight of 0.3 each. Figure 3.4 gives an illustration of the architecture of the pretext task model. For individual losses, cloud segmentation and land cover labels are categorical variables. Therefore, we used Cross-Entropy Loss [38] for them. Whereas, for DEM and S1, we used Mean Squared Error (MSE) [2] because they are continuous variables.

$$\mathcal{L}_{total} = \lambda_{seg}\mathcal{L}_{CE}(\hat{y}_{seg}, y_{seg}) + \lambda_{lc}\mathcal{L}_{CE}(\hat{y}_{lc}, y_{lc}) + \lambda_{dem}\mathcal{L}_{MSE}(\hat{y}_{dem}, y_{dem}) + \lambda_{sar}\mathcal{L}_{MSE}(\hat{y}_{sar}, y_{sar}) \tag{3.1}$$

Where:

- $\mathcal{L}_{CE}$ and $\mathcal{L}_{MSE}$ denote the Cross-Entropy and Mean Squared Error loss functions, respectively.
- $\hat{y}_i$ and y_i represent the model prediction and ground truth for task i.
- The weighting hyperparameters are set to $\lambda_{seg} = 1.0$ for the primary segmentation task, and $\lambda_{lc} = \lambda_{dem} = \lambda_{sar} = 0.3$ for the auxiliary tasks.

Finally, all the described models, their input modalities, fusion strategies, and Learning objects are neatly represented in the Table 3.2.2 below. Here, "Aux" refers to all auxiliary views, i.e., DEM, Land cover, Water Occurrence, and Azimuth.

Model Name	Input Modalities	Fusion Strategy	Learning Objectives (Outputs)
Baseline Fusion	S2 + S1 + Aux	Concatenation	Cloud Segmentation Mask (4 classes)
Gated Fusion	S2 + S1 + Aux	Gated Fusion	Cloud Segmentation Mask (4 classes)
Pretext Task	S2 Only	None (Single Encoder)	1. Cloud Segmentation Mask 2. Land Cover Classification 3. DEM Regression 4. S1 (SAR) Reconstruction

Table 3.1: Summary of Proposed Model Architectures. **S2**: Sentinel-2 (Optical), **S1**: Sentinel-1 (SAR), **Aux**: DEM, Land Cover, Water Occurrence.

Chapter 4
Experimentation

4.1 Exploratory Data Analysis

Before we dive deep into the training procedure, we analyse the dataset's statistical properties and feature distributions. This allows us to understand the dataset's intricate details and helps us better investigate its shortcomings.

4.1.1 Class Distribution and Imbalance

The CloudSen12 [5] dataset annotates four classes: thick clouds, thin clouds, cloud shadows, and background pixels. Figure 4.1 demonstrates the pixel-wise distribution of classes in the training dataset. Here, class 0 refers to background, class 1 thick clouds, class 2 thin clouds, and class 3 is shadows. One can immediately notice how the background/clear class dominates the pixel counts. Class 1 or thick clouds are second with around half the amount of pixels as that of the background, and class 2 (thin clouds) and class 3 (cloud shadows) are a distant third with around the same number of pixels. This trend is also observed in the validation and test datasets.

On the other hand, Figure 4.2 shows that the entire dataset has an equal distribution of images across varying levels of cloudiness. Here, the percentage of clouds in the image is as follows: cloud-free (0%), almost-clear (0–25%), low-cloudy (25–45%), mid-cloudy (45–65%), and cloudy (>65%). Therefore, one can conclude that even though the number of cloud and cloud-shadow pixels is much lower than that of background or cloud-free pixels, the equal distribution of images with different cloud percentages makes the dataset balanced.

4.1.2 Spatial, Temporal and Geographical Distribution of Data

A critical challenge in Earth observation, especially in multi-view learning, is the spatial and temporal distribution of satellite data. Spatial resolution is the max-

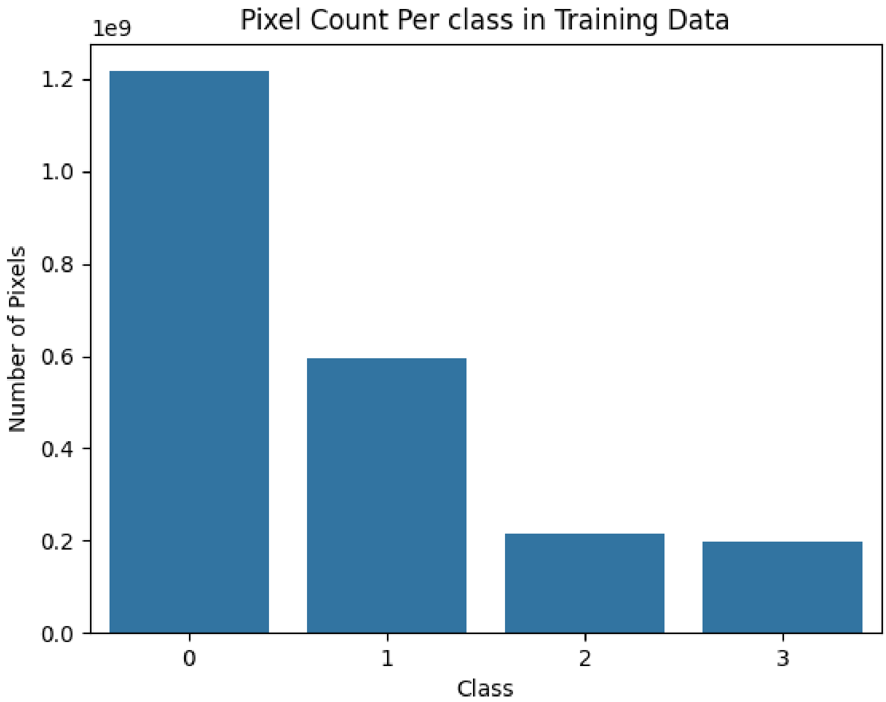

Figure 4.1: Pixelwise Distribution of Classes in the CloudSen12 training dataset

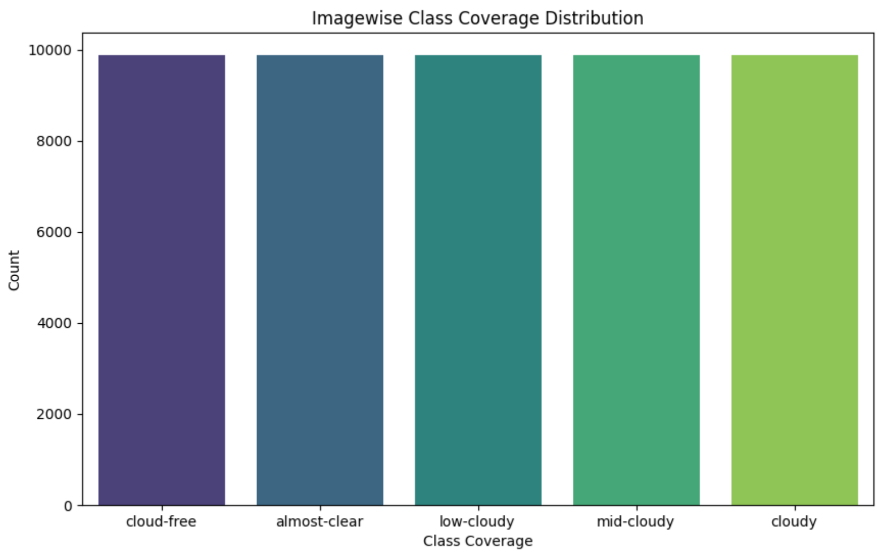

Figure 4.2: Imagewise Distribution of Classes in the CloudSen12 training dataset

imum resolution per pixel as measured by a satellite sensor. In the dataset, we use optical imagery from the Sentinel-2 satellite, which has multiple sensors with varying spatial resolutions. Typically, the Sentinel-2 satellite has spectral bands numbered 2, 3, 4, and 8 at 10m resolution. Bands 8A, 11 and 12 in 20m resolution and band 9 in 60m resolution. The authors of CloudSen12 [5] resampled each band to the highest resolution, i.e., 10 meters, so that each pixel covers a 10 m x 10 m area. Subsequently, every image in the dataset was originally 509x509 pixels, corresponding to a selected Region of Interest (ROI) with a spatial resolution of 5090x5090 meters. For our task, we resampled the images to 512x512 pixels. This was done to facilitate easier convolution and pooling during model training. Detailed information about the assets available in the CloudSen12 dataset is provided in Figure 4.3.

File/ Folder	Name	Scale	Wavelength	Description
S2L1C & S2L2A	B1	0.0001	443.9 nm (S2A)/442.3 nm (S2B)	Aerosols.
	B2	0.0001	496.6 nm (S2A)/492.1 nm (S2B)	Blue.
	B3	0.0001	560 nm (S2A)/559 nm (S2B)	Green.
	B4	0.0001	664.5 nm (S2A)/665 nm (S2B)	Red.
	B5	0.0001	703.9 nm (S2A)/703.8 nm (S2B)	Red Edge 1.
	B6	0.0001	740.2 nm (S2A)/739.1 nm (S2B)	Red Edge 2.
	B7	0.0001	782.5 nm (S2A)/779.7 nm (S2B)	Red Edge 3.
	B8	0.0001	835.1 nm (S2A)/833 nm (S2B)	NIR.
	B8A	0.0001	864.8 nm (S2A)/864 nm (S2B)	Red Edge 4.
	B9	0.0001	945 nm (S2A)/943.2 nm (S2B)	Water vapor.
	B11	0.0001	1613.7 nm (S2A)/1610.4 nm (S2B)	SWIR 1.
	B12	0.0001	2202.4 nm (S2A)/2185.7 nm (S2B)	SWIR 2.
S2L1C	B10	0.0001	1373.5 nm (S2A)/1376.9 nm (S2B)	Cirrus.
S2L2A	AOT	0.001	—	Aerosol Optical Thickness.
	WVP	0.001	—	Water Vapor Pressure.
	TCI_R	1	—	True Color Image, Red.
	TCI_G	1	—	True Color Image, Green.
	TCI_B	1	—	True Color Image, Blue.
S1	VV	1	5.405 GHz	Dual-band cross-polarization, vertical transmit/ horizontal receive.
	VH	1	5.405 GHz	Single co-polarization, vertical transmit/vertical receive.
	angle	1	—	Incidence angle generated by interpolating the 'incidenceAngle' property.
extra/	CDI	0.0001	—	Cloud Displacement Index .
	Shwdirection	0.01	—	Azimuth. Values range from 0°–360°
	elevation	1	—	Elevation in meters. Obtained from MERIT Hydro datasets
	ocurrence	1	—	JRC Global Surface Water . The frequency with which water was present.
	LC10	1	—	ESA WorldCover 10 m v100 product.

Figure 4.3: Table describing the different types of information available in Cloudsen12 [5] Dataset

But high spatial resolution is not helpful if our multi-view data is not temporally aligned. Sentinel-1 and Sentinel-2 each have different revisit times, i.e. the interval between the satellite mapping the same region twice. Further, clouds are moving objects and do not remain at a particular place for more than 24 hours. Therefore, to utilise multi-view learning across different satellites, it is imperative that their temporal resolutions differ by only a small amount. In order to investigate how far apart every Sentinel-1 image is from Sentinel-2, we plot the temporal gap between them as shown in Figure 4.4. This shows that the CloudSen12 [5] dataset is almost aligned. The images have a mismatch of $\pm$ 1 or $\pm$ 2 days with a mean gap of 0.17 days and a median gap of 1.2 days. Since there are 0 images in the dataset that are exactly temporally aligned, Sentinel-1's Synthetic Aperture Radar (SAR) provides time-invariant observations that accurately map ground features, helping distinguish them from transient atmospheric features.

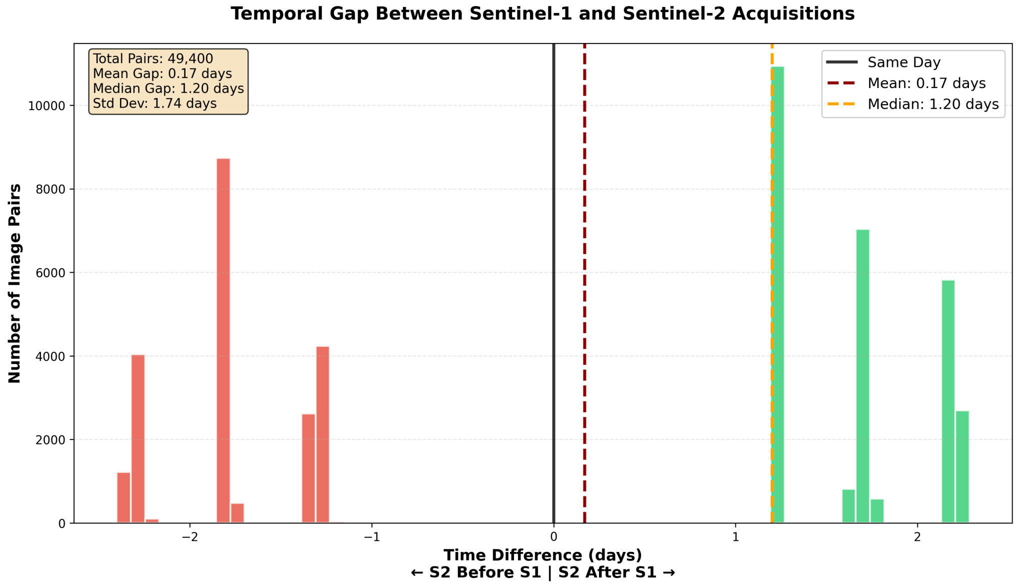

Figure 4.4: Temporal Difference in Image Acquisition between Sentinel-1 and Sentinel-2

Further, to ensure that the trained model is robust and generalises well to unseen data, it is essential that the dataset encompasses a wide range of ecosystems and atmospheric conditions. A model trained only on tropical regions will fail to detect clouds over snow-covered tundra, mountains, or glaciers due to the phenomenon known as spectral-domain shift. Figure 4.5 visualises the centroids of the Regions of Interest (ROIs) given in the dataset. According to the authors, there are 1827 unique Regions of Interest (ROI)s in the training set and 173 unique ROIs in the test set. Both the training and test datasets have equal geographic distributions of their images. This spatial diversity of the images minimises the risk of the trained model overfitting to region-specific background textures and encourages the learning of robust, globally applicable cloud features.

4.2 Training Procedure

Building on the characteristics and challenges identified in the previous section, this section describes the experimental framework used to train the single-view (baseline) and multi-view cloud and cloud shadow segmentation models. The following subsections outline the overall Model Framework, the Data Preprocessing steps undertaken to prepare the dataset for training, the Training Specifications, i.e., the hyperparameters used in the experiments, and finally the Evaluation Metrics specifically chosen to evaluate and assess the trained models' performance.

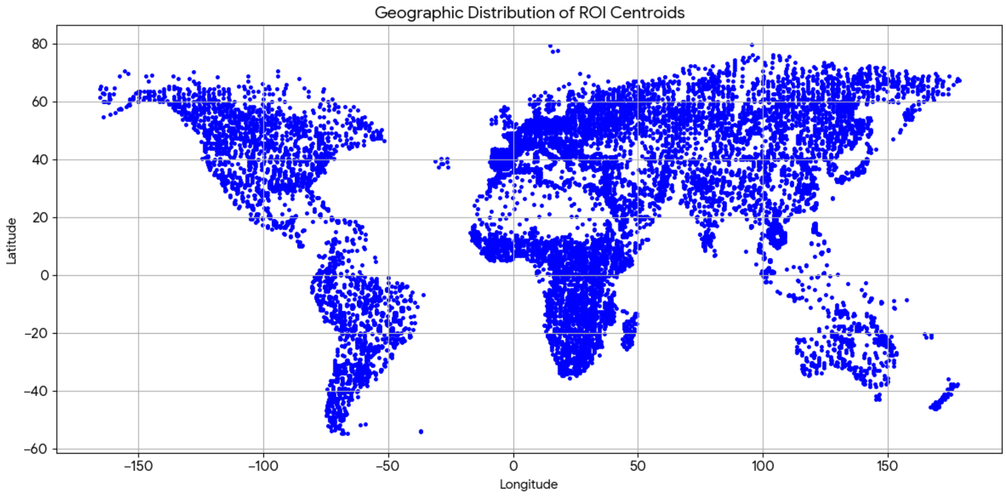

Figure 4.5: Geographical distribution of Regions of Interest(ROI) centroids

4.2.1 Model Framework

All the models were implemented in Python and use the PyTorch deep learning library. To facilitate model training, we also used PyTorch Lightning [15], which simplifies writing the deep learning pipeline. PyTorch Lightning provides built-in modules and classes to set up the training, validation, and testing regimen. Along with that, it also includes metrics that can be imported and linked to the pipeline, calculated every epoch. All these experiments were then easily tracked by using MLFLOW [9] MLOps framework, which sets up a local server to keep track of each experiment and its constituent hyperparameters.

4.2.2 Data Preprocessing

To prepare the Cloudsen12 dataset for training, Sentinel-2, Sentinel-1, and other view information were normalised using min-max scaling. Firstly, the global maximum and minimum values for all views are calculated. After that, every data point is scaled to the range of [-1,1] using the formula given in equations 4.1 and 4.2.

$$\text{View\_Std} = \frac{\text{View} - \text{Global\_View\_Min}}{\text{Global\_View\_Max} - \text{Global\_View\_Min}} \tag{4.1}$$

$$\text{View\_Normalized} = \text{View\_Std} \cdot (\text{Range\_Max} - \text{Range\_Min}) + \text{Range\_Min} \tag{4.2}$$

This is especially necessary for multi-view fusion, as it establishes a common scale for the deep learning model to integrate Sentinel-1 and Sentinel-2 information more effectively. Normalisation of data also helps deep learning models converge faster with gradient-based optimisation. For the Pretext Prediction task, the land-cover labels were encoded as one-hot vectors to make it easier for the model to distinguish the categorical labels and predict the correct landcover information.

During training, random rotations of 90, 180, and 270 degrees were used as a data augmentation technique at both the input and output levels. Lastly, the whole dataset was split into training, validation, and test sets in the ratios 85%, 5%, and 10%, respectively. This results in 8490 images per modality in the training set, 535 in the validation set, and 975 in the test set.

4.2.3 Training Specifications

Every model was trained for 50 epochs with Early stopping, which prevented the models from training beyond what was required. ADAM [24] optimiser was used as an optimiser with a weight decay [34] of 1e-5 to have a smoother learning procedure. The learning rate for all the models was set at 0.0001 with a batch size of 64. A higher learning rate made the validation loss jump up and down, signifying that the model overfits to the training set. Since our main task was generating a segmentation mask, the loss function used was Cross-Entropy Loss [38]. Whereas, in the Pretext task model, I used either Cross-Entropy Loss [38] or Mean Squared Error Loss [2] depending on the output. This is because a Cloud/Cloud shadow Segmentation mask and Landcover Product are discrete values that require a categorical-based Loss, but on the other hand, DEM and Water Occurrence[28] are continuous outputs which require a Euclidean-based Loss.

To ensure that the models are converging and they generalise well to the data, extensive ablation studies were conducted to determine the best hyperparameters. We evaluated the impact of varying dropout rates, conducted batch size experiments with gradient accumulation, and also tested the learning rates on the All Modality Gated (AMG) model. Furthermore, we compared Cross-Entropy Loss [38] with Dice Loss [**?**]. We found out that Cross-Entropy Loss yielded significantly higher F1 scores (0.69 vs 0.53) despite Dice Loss showing lower validation loss values. All the ablation studies and their results are provided in the Appendix A.

4.2.4 Evaluation Metrics

Model performances were evaluated using common metrics from the literature, like Accuracy, Precision, Recall and F1-Score. MeanIOU was also added as a segmentation task-specific metric to check for segmentation mask overlap. All the above-mentioned metrics rely on the computation of a confusion matrix, which consists of the number of samples that are either True Positive(TP), False Positive(FP), True Negative(TN) and False Negative(FN). In a binary classification scenario, a model can only predict one of two outcomes, aka positive or negative. Whereas in our case of a multi-class problem, these metrics are defined on a per-class basis. Here, True Positive refers to those predictions where the model correctly predicts the desired class Alpha, and it is indeed class Alpha. A False Positive occurs when the model incorrectly predicts a class as Alpha when it is a different class in reality. A True Negative occurs when a model correctly predicts the class as Not Alpha, and it is indeed Not Alpha in reality. Finally, a False Negative Scenario occurs when a model predicts a class as Not Alpha, but it is class Alpha in reality. These metrics are calculated per class and then averaged over all the classes to get the results for the overall model. The accuracy metric in an image segmentation task is defined as the number of pixels that are correctly classified with respect to the total number of pixels.

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FN} + \text{FP}} \tag{4.3}$$

Precision refers to the proportion of pixels that are correctly labelled to the total number of predicted pixels in an image.

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{4.4}$$

Recall, or True Positive Rate, is similar to Precision in that it is defined as the proportion of correctly labelled pixels to the sum of correctly predicted pixels and false negative pixels. Therefore, recall measures the number of correctly labelled pixels in an image.

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{4.5}$$

F1 Score or Dice coefficient is the harmonic mean of precision and recall. In segmentation tasks, it is defined as the proportion of twice the area of overlap to the total area i.e. the sum of predicted samples and ground truth samples. F1 Score is properly defined in equation 4.6 given below.

$$\text{F1 Score} = 2 * \frac{\text{Precision} * \text{Recall}}{\text{Precision} + \text{Recall}} \tag{4.6}$$

Finally, the Intersection over Union (IOU) score measures the ratio of the Intersection of the predicted bounding box and the ground truth bounding box with respect to the Union of the two mentioned bounding boxes. The mean Intersection over Union (mIOU) is the average of the IOU scores across all classes. We used mIOU scores from equation 4.8 to evaluate the performance of our models.

$$\text{IoU} = \frac{\text{Intersection}}{\text{Union}} \tag{4.7}$$

$$\text{mIOU} = \frac{1}{N} \sum_{i=1}^{N} \frac{\text{TP}_i}{\text{TP}_i + \text{FP}_i + \text{FN}_i} \tag{4.8}$$

4.3 Results

4.3.1 Competing Models

In this section we share the proposed model's performance as compared with other baseline models. The first set of baseline models are traditional algorithm based models for cloud and cloud shadow segmentation i.e. Sen2Cor [37] and KappaMask [11]. Both of these algorithms utilize Sentinel-2 optical satellite infomation as input data for their models. Further, these models were selected because their output classes correspond to the four afformentioned classes in CloudSen12 [5] dataset. Cloudsen12 [5] already provides KappaMask [11] and Sen2Cor [37] labels which were compared with the high quality manually annotated labels for comparison. Moving on to deep learning models, two single-view UNet [47] models were constructed for Sentinel-1 and Sentinel-2 information respectively. These models have one encoder and one decoder each. The four deep learning models were trained using the same training specifications given in the previous subsection. Table 4.1 illustrates the list of all baseline models and their performance on the aforementioned evaluation metrics.

Model	Fusion	Accuracy	Precision	Recall	F1Score
Sen2Cor	-	0.49	0.59	0.49	0.45
KappaMask	-	0.52	0.58	0.52	0.50
Sentinel 1 Only	-	0.25	0.13	0.25	0.17
Sentinel 2 Only	-	0.65	0.70	0.65	0.63

Table 4.1: Table comparing the baseline models' performances

4.3.2 Multi-view Models

A significant development from single-view models is two dual-view fusion models comprising Sentinel-1 and Sentinel-2 information as inputs. Two models were constructed, each having two encoders for Sentinel-1 and Sentinel-2 information, which are fused at the bottleneck layer. The cloud and cloud shadow segmentation output is achieved using one decoder, respectively, for each model, which uses the fused information as input. The only defining difference between the above two models is the fusion technique used. One model consists of concatenation fusion, and the other consists of gated fusion [4]. These models will subsequently be referred to as **"2 Modality Concat (2MC)"** and **"2 Modality Gated (2MG)"**. Subsequently, two more models were constructed, which combined not just Sentinel-1 and Sentinel-2 information, but also DEM [55], Landcover [57], Water Occurrence [44] and Azimuth information. These models also consist of individual encoders for every view, whereas a single decoder is used for the output. The above two models are referred to as **"All Modality Concat (AMC)"** and **"All Modality Gated (AMG)"**. Table 4.2 illustrates their performance using the evaluation metrics mentioned in the previous section.

Model	Fusion	Accuracy	Precision	Recall	F1Score
2 Modality	Concatenation	0.67	0.72	0.67	0.69
2 Modality	Gated	0.66	0.72	0.66	0.66
All Modality	Concatenation	**0.69**	**0.77**	**0.69**	**0.72**
All Modality	Gated	0.66	0.74	0.66	0.68

Table 4.2: Table comparing the multi-view models' performances

4.3.3 Pretext Task Model

The final model that was constructed was a "Pretext task" model. Its purpose was to check if a model trained on optical information from Sentinel-2 can be used for more than one downstream task. Therefore, this model consists of 1 input encoder with Sentinel-2 optical information and has 4 decoders, each for one specific output task. The 4 output tasks are cloud and cloud shadow segmentation, elevation maps, land use land cover classification and finally SAR. Due to limitations of time, this model could not be fully tested. As a result, the only comparison that was available was to check the performance of the "Pretext Task" Model on cloud and cloud shadow segmentation with the baseline model constructed using Sentinel-2 only. As evident from Table 4.3, we see that the Pretext Task model is worse in comparison with the Sentinel-2 only baseline model. One possible reason for this might be the addition of the extra output information in the decoders. This is because, when the weights of the model are updated after backpropagation, it takes into account the false classifications in every decoder and, in turn, messes up the weights.

Model	Accuracy	Precision	Recall	F1Score
Sentinel 2 Pretext	0.61	0.60	0.67	0.61
Sentinel 2 Only	0.65	0.70	0.65	0.63

Table 4.3: Table comparing the Pretext-task model's performance to the Sentinel-2 baseline on Cloud segmentation task

4.3.4 Quantitative Results

Traditional algorithmic models i.e. Kappa-Mask [11] and Sen2Cor [37] achieve around 50% performance across all metrics. Further, table **??** shows that using just Sentinel-1 Synthetic Aperture Radar (SAR) information for a deep learning model has inferior performance than even traditional non-deep learning algorithms. Performance starts increasing drastically as we begin using Sentinel-2 optical information. Combining Sentinel-1 and Sentinel-2 information provides a marginal improvement in overall performance. This suggests that, though not as powerful alone, Sentinel-1 does help detect cloud and cloud shadow boundaries when combined with optical information. Whereas, combining DEM [55], Landcover [57], Water Occurrence [44], and Azimuth information along with Sentinel-1 and Sentinel-2 information improves cloud and cloud shadow detection capabilities by a significant margin. Concatenation Fusion also comes up as a clear winner in quantitative performance as compared to Gated Fusion [4]. Comparing the models All Modality Gated (AMG) versus 2 Modality Gated (2MG) in table **??**, we observe that the AMG model shows improved performance metrics in Precision and F1 Score. But on the other hand, the Accuracy and Recall score remain exactly the same. On the other hand, the All Modality Concat (AMC) model shows significant improvement over its counterpart 2 Modality Concat(2MC) in all performance metrics. One hypothesis for this result might be because of the way gated fusion works. Gated fusion intelligently prioritises one set of features over the other by assigning higher weights to more impactful features and lower weights to others. Whereas in concatenation fusion, all features get equal importance as they get stacked on top of each other at the bottleneck layer.

4.3.5 Robustness Results

Cloudsen12 [5] provides us with metadata information for every data sample, such as the annotator's names, landcover labels, difficulty scores, Sentinel-1 and Sentinel-2 acquisition date, cloud coverage, coordinates of the image in latitude and longitude, among others. Difficulty scores denote the manual annotator's confidence in labelling images on a scale from 1 to 5; 1 being high confidence samples, which are easy to annotate, and 5 implies hard-to-annotate samples. Cloud coverage, as mentioned in previous chapters, indicates the percentage of the image covered in clouds; from cloud-free (0% clouds) to cloudy (>65% clouds). Landcover label shows the predominant land use in the image using the ESA WorldCover [57] 10m scale with categorical labels ranging from 10 to 100. To make it legible for human understanding, the categorical labels were converted to their corresponding land use labels such as Tree Cover, Shrubland, Snow and Ice. The centroid coordinates were used to find the corresponding Koeppen-Geiger Climatic Zones [25]. The KG climate zones are a system of classifying the Earth's surface according to the relative heat and humidity it receives throughout the year. The KG climate zones are divided into 5 broad categories, namely: A - Tropical, B - Arid, C - Temperate, D - Continental, E - Polar and finally, water bodies are classified as 0 - Ocean.

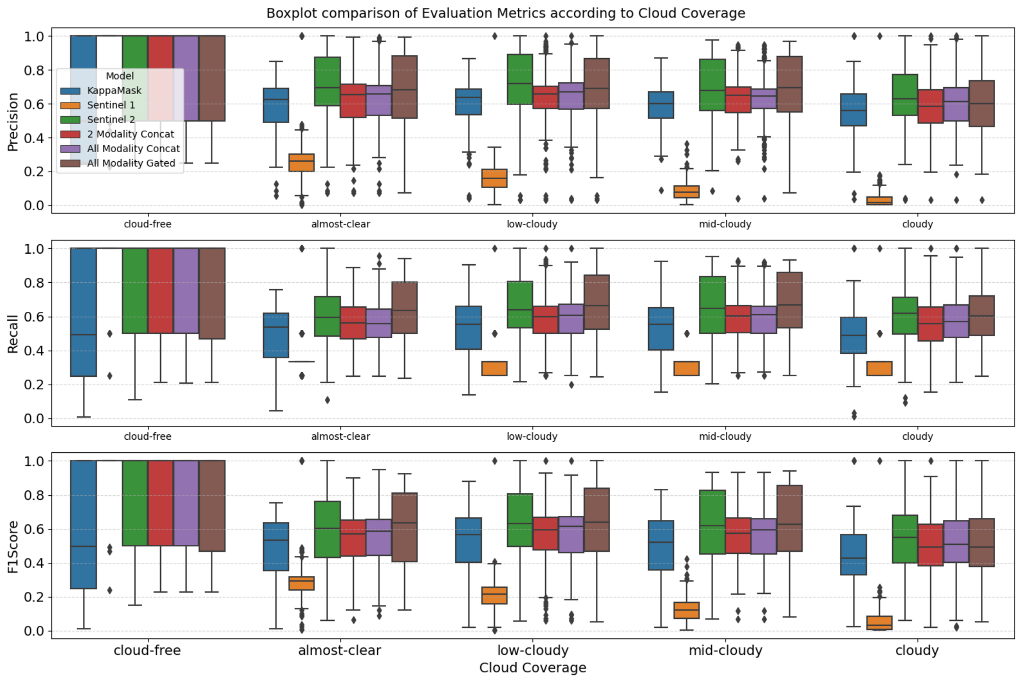

Figure 4.6: **Box plot comparison of six models' performance according to cloud coverage.** The evalution metrics are illustrated on the Y axis and the cloud coverage in ascending order on X axis.

Figure 4.6 shows how the All Modality Gated (AMG) model beats other competing models in the difficult-to-classify cloud coverage conditions, such as almost-clear and low-cloudy. These conditions have thin clouds present, which make it difficult for optical satellites to detect. The AMG model scores significantly higher in Recall and F1Score metrics against All Modality Concat (AMC) and 2 Modality Concat (2MC). Yet, it scored strikingly similar to the baseline model constructed on just Sentinel-2 optical information. In cloudy conditions, the Sentinel-2 baseline model beats all others and has significantly fewer outliers.

Figure 4.7 illustrates how the different models perform on increasing order of difficulty. The AMG model is able to achieve significantly better results than the

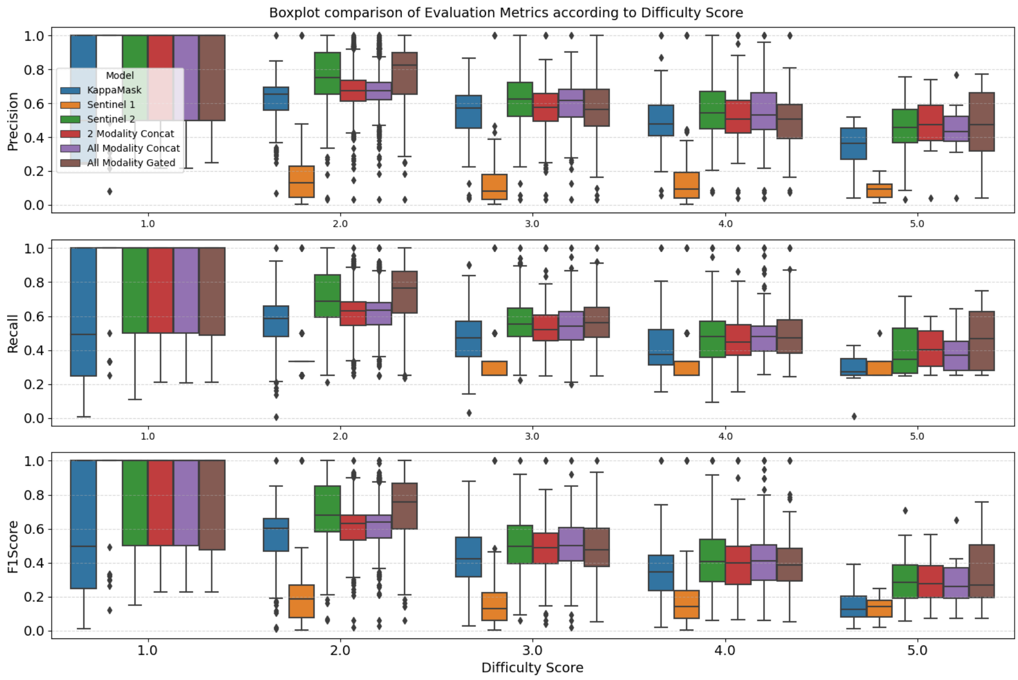

Figure 4.7: **Box plot comparison of six models' performance according to Difficulty scores.** The evalution metrics are illustrated on the Y axis and the difficulty score in ascending order on X axis.

competing models in low-medium difficulty images. Various outliers are also observed in the competing models of 2 Modality Gated (2MG) and All Modality Concat (AMC). Whereas, in medium-high difficulty samples of **3.0** to **4.0**, the AMG model achieves equal or even lower performance with competing gated and concatenation models. Lastly, in very high, aka **5.0**, difficulty, the AMG model again achieves better performance, as evident from the Precision and Recall scores. The All Modality Concat (AMC) model always scores lower than or equal to the AMG and Sentinel-2 baseline model. It also has the largest number of outliers across different difficulty scores and performance metrics. On the other hand, the AMG model has the fewest number of outliers.

Figure 4.8 describes the major models' F1Score distribution over different KG climate zones [25]. The All Modality Gated (AMG) model performs better in polar climates that are referred to as **E** in the accompanying image. This points to the model's ability to distinguish between clouds and snow significantly better than other competing models. The AMG model also shows better F1Score in temperate climate conditions referred to as D in the image. Whereas, in all other climate zones, the AMG model performs better than the 2 Modality Concat (2MC) and All Modality Concat (AMC) models but fails to beat the single-view Sentinel-2 model. In Oceanic regions, aka "O", we can see that all combinations of S1 and S2 manage to achieve better F1Score than using Sentinel-2 alone. This might reinforce the fact that using Sentinel-1's radar information is important in order to distinguish between the reflection of water and clouds.

Figure 4.9 illustrates the F1Score for ten landcover types. The AMG model is able to

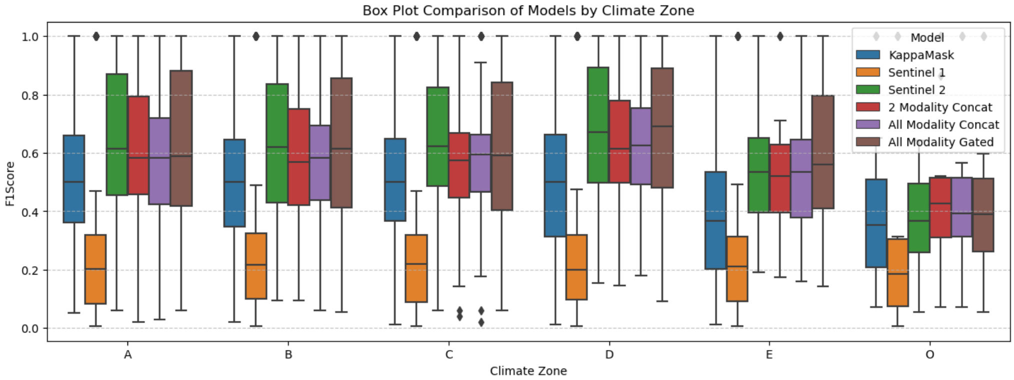

Figure 4.8: **Box plot comparison of six models' F1Score according to Koeppen-Geiger Climatic Zones [25].** The F1Score is illustrated on the Y axis and the climate zone on the X axis.

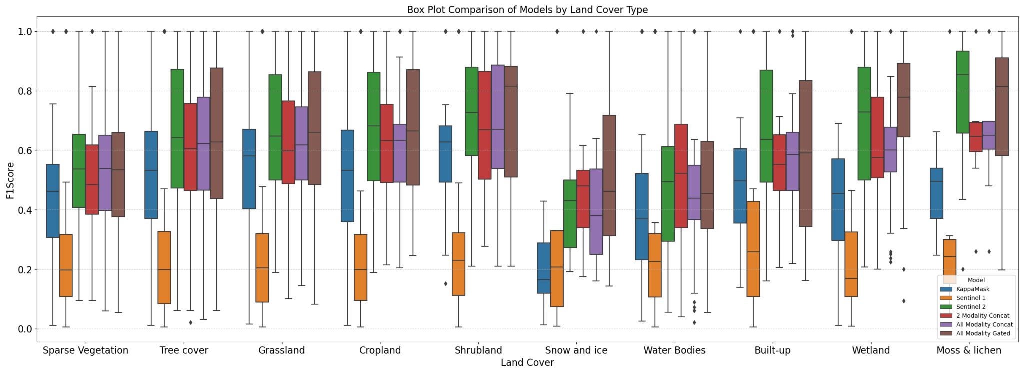

Figure 4.9: **Box plot comparison of six models' F1Score according to Land Cover.** F1Score is illustrated on the Y axis and the landcover label on the X axis.

outperform all the competing models in snow and ice conditions. In wetlands, the AMG model shows much higher confidence in cloud/cloud shadow predictions than other competing models. Whereas, in the case of built-up areas, the AMG model shows the least amount of confidence, i.e. a varied set of predicted labels. In Moss and lichen landcover types, the AMG model beats the AMC and 2MC models but fails to beat the Sentinel-2 baseline model. Further, in water body landcover, the 2MC model is able to beat others in F1Score. In all other landcover types, the AMG model performs better than other competing multi-view models. But its performance is marginally better or equal to the baseline Sentinel-2 model.

In summary, we observe that despite not being the overall best model according to Table **??**, the All Modality Gated (AMG) model is the most stable and consistent model across different cloud coverages and difficulty scores. It is also able to distinguish much better between clouds and snow. Whereas the All Modality Concat (AMC), despite being the overall winner in performance metrics, shows poor performance in the above scenarios and also consists of a considerable number of outliers.

4.3.6 Band Removal Experiment

In order to further test the robustness of the all modality models, aka AMC and AMG, we decided to simulate a data loss experiment. In such a scenario, the Sentinel-2's (S-2) bands stop functioning one after the other. In order to simulate this, we replaced the S-2 Band values in the Test set to "0" starting from Band 1 till Band 12. To keep the simulation simple, we started from Band 1 and consecutively increased the number of Bands with value 0 in every turn. Those results are illustrated in the table 4.4 and 4.5 below.

Across both models, one can observe that there is an immediate drop in performance when the fist band aka Band 1 (Aerosols) is removed and it remains constant in performance till Band 2(Blue) is removed. The performance drops staggeringly as soon as Band 3(Green) is removed, and then remains almost stagnant until Band 5 is removed. Model Performance seems to jump back, which signifies that the AMG model needs the combination of Blue, Green and Red together to give tangible results. If two of them are removed, then the model loses accuracy in its predictions. Performance drops again when Near Infrared (NIR) Band 8 is removed. The last sudden increase in performance happens after Band 10's removal. In such a case, only Short Wave Infrared (SWIR) and Cirrus bands of S-2 are capable enough to give as close a performance as when only 1 band is removed. This experiment suggests that the AMG model is only robust in certain combinations of bands.

Bands Removed	Band Name	Accuracy	Precision	Recall	F1Score	MeanIOU
0	-	0.66	0.74	0.66	0.68	0.41
1	Aerosol	0.45	0.46	0.45	0.44	0.25
2	Blue	0.45	0.46	0.45	0.44	0.25
3	Green	0.29	0.33	0.29	0.20	0.28
4	Red	0.25	0.32	0.25	0.11	0.28
5	Red Edge 1	0.25	0.32	0.25	0.11	0.28
6	Red Edge 2	0.45	0.43	0.45	0.40	0.22
7	Red Edge 3	0.49	0.53	0.49	0.41	0.31
8	NIR	0.34	0.32	0.34	0.27	0.28
9	Red Edge 4	0.26	0.37	0.26	0.13	0.29
10	Water Vapour	0.43	0.36	0.43	0.32	0.31
11	Cirrus	0.39	0.45	0.39	0.37	0.28
12	SWIR 1	0.25	0.34	0.25	0.12	0.28

Table 4.4: Simulated Band Loss Scenario in All Modality Gated Fusion Model

On the other hand, in the case of the All Modality Concat (AMC) model, we see a sharper decline in performance metrics as soon as the first band, i.e., the aerosol band, is removed. The Accuracy, Precision and Recall metrics immediately become half of their original performance. Whereas the F1Score drops to around one-third of the original. The MeanIOU, on the other hand, decreases by a third. Keeping in line with the above test, accuracy drops more by the time Band 3 (Green) is removed. Similarly, the performance jumps back when band 6 (Red Edge 2) is removed. Here, the increase in performance is much more than that observed in the AMG model. The AMC model with 7 bands removed performs better than the AMC model with just 1 band removed. Unlike the AMG model, as soon as Band 9 (Red Edge 4) is removed, the model shows an increase in performance. After that, the decline continues until Band 11 (SWIR 1) is removed, when another slight increase in performance is observed. After 12 bands are removed, the AMC model has a slightly higher performance than when all 12 bands are removed in the AMG model above. The exact metrics are given below in Table 4.5.

Bands Removed	Band Name	Accuracy	Precision	Recall	F1Score	MeanIOU
0	-	0.69	0.77	0.69	0.72	0.43
1	Aerosol	0.31	0.35	0.31	0.20	0.31
2	Blue	0.31	0.35	0.31	0.20	0.31
3	Green	0.25	0.31	0.25	0.11	0.28
4	Red	0.25	0.31	0.25	0.11	0.28
5	Red Edge 1	0.25	0.31	0.25	0.11	0.28
6	Red Edge 2	0.39	0.34	0.39	0.24	0.31
7	Red Edge 3	0.40	0.39	0.40	0.22	0.33
8	NIR	0.25	0.33	0.25	0.12	0.28
9	Red Edge 4	0.41	0.43	0.41	0.22	0.34
10	Water Vapour	0.37	0.60	0.37	0.30	0.31
11	Cirrus	0.26	0.27	0.26	0.14	0.28
12	SWIR 1	0.30	0.50	0.30	0.22	0.30

Table 4.5: Simulated Band Loss Scenario in All Modality Concatenation Fusion Model

Some plausible hypotheses for this "dip and recovery" phenomenon observed in the band loss experiments are as follows:

(a) **Corrupted Feature vs Missing Feature Hypothesis**:

- Convolutional Neural Networks (CNNs) learn the underlying spectral signatures of optical satellite images. In the Visible and Near-Infrared (NIR) spectrum, clouds typically appear as bright features.
- **The Dip (Bands 1-5 Removed)**: When the visible spectrum is "removed", i.e. their values are zeroed out, we create a physically impossible spectral signature that is completely black in visible but bright in the NIR. The model interprets this input as highly noisy or a very confusing class, which ultimately leads to massive missclassifications and therefore lower accuracy.
- **The Recovery**: When both the Red Edge and NIR bands are removed, a much wider spectrum becomes black. This uniform darkness across the optical spectrum most likely prompts the model to shift towards either classifying everything as background. Or it shifts its reliance from Sentinel-2 optical data to a "safer" Sentinel-1 SAR, DEM and other auxiliary features, which are theoretically still intact.
- According to Hendrycks et al. [21], when a model encounters test cases outside its distribution, it becomes confused. This Out-of-Distribution (OOD) issue is further exacerbated by the "conflicting" state being farther from the training distribution than the "empty" zeroed-out state.

(b) **Intelligent vs Naive Fusion of Information**:

- In the All Modality Concat (AMC) model, all views are mixed equally and given equal importance. Initially, when only a few Sentinel-2 bands are zeroed out, the model misclassifies due to the confusing input information discussed above. But when enough S2 bands are zeroed out, the model gets dominated by Sentinel-1 SAR data. Thus, S1 features dominate decision-making, leading to lower performance.

- In the All Modality Gated (AMG) method, each view is assigned its own weight by the Gated Fusion function. However, these weights are trained on valid data without any band loss scenarios. Therefore, initially, AMG follows the same pattern as AMC, but once enough S2 bands are zeroed out, it does not naively give importance to S1. Rather, other secondary features, such as land cover and DEM, are given higher priority, making it more robust than the AMC model.
- This phenomenon is also observed in Baltruschat et al. [6], who remove textual information from their multimodal medical image classification model and rely solely on image data for final classification. Their model fails to produce accurate results because it also uses feature-level fusion with a concatenation function at the bottleneck.

(c) **Significance of the Red Edge**:

- According to Drusch et al. [13], the Sentinel-2 optical satellite has multiple Red Edge bands specifically designed to detect vegetation reflectance associated with chlorophyll. Clouds do not have a Red Edge and are flatly bright. On the other hand, the background vegetation does indeed have a Red Edge.
- When the visible Red Band (Band 4) is removed, but the Red Edge (Bands 5, 6, and 7) is kept intact, the models hallucinate and classify cloud pixels as Background pixels. This is because there is a steep red edge from 0 in Band 4 to high in Band 5 due to "information loss".
- When the red edge bands are also removed, this false steep red-edge signal is removed, and the model reverts to Sentinel-1 or other auxiliary information, stopping in some cases from falsely predicting clouds as background.

Chapter 5
Conclusion

5.1 Summary

Remote Sensing plays a crucial role in monitoring the Earth's surface, yet its effectiveness is often hindered by atmospheric conditions such as clouds and cloud shadows. This problem is exacerbated when we have access only to optical satellite imagery. With clouds covering approximately 67% of the Earth's surface at any given time, as measured by King et al.[23], optical satellites suffer frequent occlusion from data, and this in turn hampers downstream tasks such as Land Use Land Cover and deforestation analysis because it makes a lot of the data unusable. Traditional cloud and cloud shadow segmentation algorithms relied solely on features learned from optical images, such as thresholding, leading to critical failures in polar regions and a lack of resilience when sensor bands failed. This thesis aimed to address these limitations by constructing and investigating a multi-view deep learning model that utilised radar and other non-optical data to resolve these spectral obscurities.

5.1.1 Model Proposal and Architecture

To test the advantages of multi-view learning, we proposed and implemented a Multi-View U-Net [47] architecture. This architecture uses distinct encoders to process different views or modalities such as Sentinel-2 Optical data, Sentinel-1 Synthetic Aperture Radar (SAR) data, Digital Elevation Models (DEMs) [55], Land Cover [57], Azimuth and Water Occurrence [44] data. We specifically investigated two fusion mechanisms at the bottleneck layer: **Concatenation Fusion**, which naively stacks all views together, and [**?**], which dynamically assigns weights to opinions based on their information value. Furthermore, an inverse U-Net architecture was proposed that was trained solely on Sentinel-2 optical imagery and aimed to investigate whether it can reconstruct geometric modalities such as DEMs, Landcover, and SAR.

5.1.2 Validation Scenario

The proposed models were rigorously tested and validated using the CloudSen12 benchmarking dataset [5]. This comprehensive dataset had global coverage and enabled evaluation across a wide range of biomes, climatic zones and cloud coverages. The validation approach utilised standard metrics, including Accuracy, Precision, Recall, and MeanIOU. But it also went further and tested these scores against Köppen-Geiger Climatic Zones [25], Land Cover types, cloud coverage percentages, and Annotation difficulty scores. Additionally, a robustness experiment was conducted that simulated sensor failure by incrementally "removing" Sentinel-2 bands by replacing their values with 0s. This assessed whether a model trained on six views could utilise secondary modalities, such as SAR and DEM, and still achieve accurate segmentation results.

5.1.3 Summary of Contributions

The principal contributions of this master's thesis are as follows:

(a) **Fusion Strategy**: While the All Modality Concat (AMC) model achieved higher performance according to the tested metrics, the All Modality Gated (AMG) model demonstrated superior consistency and stability, particularly in Polar and Wetland climatic conditions.

(b) **Spectral Disambiguation**: Integrating DEM, Landcover and other auxiliary data into the fusion strategy effectively mitigated the typical confusions faced by cloud segmentation models; i.e. distinguishing between snow and clouds. This remains a persistent cause of failure in optical-only models.

(c) **Band Loss Robustness Analysis**: The limits of the current multi-view fusion models were realised with respect to band loss. Even though the AMG model exhibited some capacity to utilise non-optical views during partial data loss, it still suffered a significant drop in performance. It was much worse in the AMC model's case. Explicit training or more complex models are necessary to prevent performance degradation in cases of sensor failure.

5.2 Evaluation of the overall Goal

The ultimate goal of this research was to explore the advantages of multi-view satellite data for cloud and cloud shadow segmentation. This goal was evaluated against the objectives defined in Chapter 1:

- **Fusion Strategy Exploration**: This objective was achieved as we observed a clear distinction between naive Concatenation Fusion and intelligent Gated Fusion [4]. The Concatenation Fusion prioritised precision in normal scenarios, whereas the Gated Fusion worked better in out-of-distribution scenarios and hard-to-segment conditions.
- **Robustness Assessment**: This objective was also fulfilled, as the thesis quantified the limits of these models. It confirmed that multi-view learning effectively addresses environmental noise such as snow and reflecting water. But it cannot address information-loss scenarios, such as missing bands, without further training.

- Pretext Task Viability: This investigation of the inverted U-net pretext task model yielded negative results; i.e. the model's cloud and cloud shadow segmentation performance degraded. This indicates that trying to learn geometric features such as SAR and DEM alongside segmentation and land cover can introduce significant complexity, thereby reducing performance. Unfortunately, due to time constraints, explicit testing of land cover classification, DEMs, and SAR reconstruction was not conducted.

5.3 Future Work

The experiments conducted in this thesis identify various high-potential avenues for future research. To fully unlock the benefits of multi-view learning, future work should move beyond standard CNN architectures and adopt more adaptive frameworks.

(a) **Transformer-Based Networks**: While this thesis utilised a CNN-based encoder-decoder architecture, recent research has shown that Vision Transformers [12] are much better at capturing the multi-modal nature of satellite data and effectively fusing them together. Alistair Francis [16] and Wang et al. [**?**] demonstrate that global semantic relationships across multiple views are captured more effectively by transformers than by CNNs. Alistair Francis's transformer network is sensor-independent and can simultaneously ingest Sentinel-2, Sentinel-1, and DEMs. Future iterations should replace the U-Net backbone with a SWIN Transformer [32] to achieve better results.

(b) **SAR Specific Pre-training**: Our results showed that Sentinel-1 SAR data were underutilised until Sentinel-2 optical data were removed during the band dropout experiment. To force the model to learn better radar features, SARMAE Liu et al. [31] could be employed. This pre-training objective targets SAR representation learning by injecting speckle noise and forcing the model to recover the clean signal, ensuring SAR becomes a strong, standalone predictor.

(c) **Masked Image Modelling**: To address the limitations faced by our models in band dropout/sensor failure scenarios, Labatie et al. [27] demonstrate how multi-modal and multi-temporal information can be used to train an earth observation model that is used to reconstruct missing information using Masked Autoencoders (MAE) [18].

(d) **Foundation Models**: Finally, rather than training individual models from scratch, future research should utilise Remote Sensing Foundation Models (RSFMs) such as AlphaEarth [7] by Google. As surveyed by Xiao et al. [53] last year, RSFMs can achieve zero-shot generalisation to challenging tasks such as cloud segmentation in polar climates and may also address the scarcity of data for certain ecosystems.

Appendix A
Supplementary Material

A.1 Ablation Studies

In this section, we present all ablation studies conducted to determine the optimal hyperparameters for our multi-view models. These experiments justify the selection of dropout rates, batch sizes, learning rates, and loss functions that are used in the main experimental framework.

Model	Dropout	Validation Loss
AMG	-	0.558
AMG	10%	0.771
AMG	20%	1.530
AMG	30%	1.512
AMG	Adaptive	**0.553**

Table A.1: Dropout Experiment

Model	Batch Size	Gradient Accumulation	Effective Batch Size	Validation Loss
AMG	16	2 Steps	32	0.553
AMG	16	4 Steps	64	0.552
AMG	16	8 Steps	128	0.961

Table A.2: Effective Batch Size Experiment with Gradient Accumulation

$$\text{Dice Score(Ytrue, Ypred)} = 2 * \frac{\text{Ytrue Ypred}}{|\,\text{Ytrue}\,| + |\,\text{Ypred}\,|} \tag{A.1}$$

$$\text{Dice Loss} = 1 - \text{Dice Score(Ytrue, Ypred)} \tag{A.2}$$

Model	Learning Rate	Validation Loss	Accuracy	Precision	Recall	F1Score
AMG	0.0001	0.497	0.674	0.749	0.674	0.694
AMG	0.001	0.552	0.610	0.751	0.610	0.647

Table A.3: Learning Rate Experiments

Model	Loss Function	Validation Loss	Accuracy	Precision	Recall	F1Score
AMG	Cross Entropy	0.497	0.674	0.749	0.674	0.694
AMG	Dice Loss	**0.216**	0.551	0.574	0.551	0.537

Table A.4: Loss Function Experiments

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